Bend and SNAP? The effect of food availability on workplace injuries

Gina Pagan

Department of Agricultural and Resource Economics, University of California, Davis

gpagan@ucdavis.edu

October 13, 2026

Abstract

I leverage exogenous variation in the timing of SNAP benefit disbursal to estimate how relative food availability affects workplace injuries. SNAP recipients spend most of their benefits in the first two weeks after disbursal, and prior research shows that this pattern, known as the SNAP cycle, affects cognitive performance and health, but its effect on worker safety has not been established. Using administrative workers' compensation claims from Oregon over 2005–2018 and a predicted measure of SNAP eligibility, I find no effect when workers are pooled, but large and offsetting effects by gender among workers aged 18 to 24. Injuries for young women are highest in the first days after benefits arrive, while injury rates for young men rise in the second half of the benefit month by 17.6 percent for non-severe and 25.4 percent for severe injuries. Because severe and non-severe injuries move proportionally, these patterns are most consistent with changes in hours worked. Among men aged 45 to 54, by contrast, the rate of severe injury rises by 47 percent on the day benefits are received while non-severe injuries do not, a shift that labor supply alone would not produce. These intent-to-treat estimates likely understate effects for SNAP participants.

1 Introduction

The Supplemental Nutrition Assistance Program (SNAP) is the largest federally-funded nutrition assistance program in the United States (US), serving 12 percent of the population in 2024 (Jones, 2025). In July 2025, the One Big Beautiful Bill Act (OBBBA) implemented a plan to cut SNAP spending by $187 billion through 2034 by extending work requirements to a broader population, shifting additional administrative costs to state departments, and eliminating eligibility for many lawful immigrants. Since then, approximately 5 million people have lost access to SNAP (Rosenbaum et al., 2026), the majority of whom are working-age adults, and these changes are likely to have consequences for food security in poor households (Cook and East, 2026).

Economists have increasingly used natural experiments to study how depletion of caloric and financial resources within a month affects outcomes for the low-income population in the US. The evidence suggests that when resources are scarce, students perform worse on tests (Gassman-Pines and Bellows, 2018; Cotti et al., 2018; Kuhn, 2018; Bond et al., 2022), there are more thefts from grocery stores (Carr and Packham, 2019), and emergency room visits increase (Seligman et al., 2014; Arteaga et al., 2018; Cotti et al., 2020). However, relatively little of this literature focuses on workers. Primarily descriptive evidence from other disciplines suggests correlations between food insecurity and workers' productivity (Moreno et al., 2025; Moy et al., 2026) and injury risk (Cubbin et al., 2000; Men et al., 2021). Development economists have found causal evidence for the former (Thomas et al., 2006; Hu and Wang, 2019), and to my knowledge the second relationship has not been studied in a US context.

As a larger population of SNAP recipients face work requirements, it is increasingly important to understand how socioeconomic conditions affect outcomes in the workplace. Occupational injuries account for nearly 40 percent of injuries experienced by working adults each year (Smith et al., 2006), and a larger fraction for men. NSC (2026) estimates that lost productivity and wages accounted for 30 percent of the total economic cost of workplace injuries ($181.4 billion) in 2024. This total cost estimate does not include the cost incurred by employers to purchase workers' compensation (WC) insurance, which totaled approximately $103 billion in 2022 (NASI, 2024)[1]. Similar to the well-documented socioeconomic health gradient, there is also evidence of a socioeconomic safety gradient. Low-wage workers have higher rates of

[1]The National Academy of Social Insurance (NASI) publishes annual reports detailing costs associated with WC insurance and the report using 2022 data is the latest available.

non-fatal injuries (Steege et al., 2014) and have higher mortality rates from injuries (Steenland et al., 2003). Furthermore, research shows that other external factors affecting cognition, like extreme temperatures (Dillender, 2021) or particulate matter exposure (Cabral and Dillender, 2024; Lee and Beatty, 2025), increase incidence of workplace injuries, so it is reasonable that workers experiencing acute food insecurity face a similar change in injury risk.

This paper aims to fill this gap in the literature by asking whether short-run food scarcity exacerbates the risk of occupational injury for low-wage workers. I explore this question in the context of Oregon SNAP participants over a 14 year period from 2005 through 2018, though ongoing work extends this analysis to other states. Oregon is an ideal setting to answer this question for two reasons: (1) the day that SNAP households receive their benefits each month is both exogenously determined and plausibly random; and (2) the SNAP participation rate among eligible working-poor households in Oregon is consistently one of the highest in the US (Cunnyngham, 2021), and exceeded 75 percent in all sample years.[2] States commonly stagger the disbursement of SNAP benefits over multiple days to minimize retailer strain and in Oregon, benefits are distributed over the first 9 calendar days of each month, and the exact day is determined by the terminal digit of the household's SNAP account holder's social security number (SSN).[3] This generates variation in the number of days since benefits were last received for otherwise similar workers on any particular day, allowing me to compare injury outcomes for workers exogenously assigned to be either early (when food is relatively accessible) or late in their SNAP benefit month. The second point matters because my injury data come from administrative workers' compensation (WC) records which do not contain data on SNAP participation. Because SNAP participants are a minority in the workforce, I use American Community Survey (ACS) data to construct a probability measure for SNAP eligibility based on information that I observe in my WC claims, namely gender, age, occupation, and average weekly wage. My analyses restrict to workers above specified thresholds of probabilistic SNAP *eligibility*, but Oregon's high participation rate among eligible households enables this to be a reasonably close approximation of probabilistic SNAP *participation*. All estimates I present should thus be interpreted as intent-to-treat (ITT) effects.

My empirical approach is motivated by the repeatedly documented phenomenon that food expenditures and consumption are not smoothed over the month between benefit disbursals for SNAP participants (Wilde

[2]The Food and Nutrition Administration (FNA) publishes annual reports with estimates of SNAP participation rates by state for all eligible households and for those where at least one member works for pay (or 'working poor' households). These reports are accessible at https://www.fna.usda.gov/research/snap/state-participation-rates.

[3]Each participating household chooses an adult to be primary account holder who is the primary point of contact for logistics and case management. I will use 'SNAP account holder' and 'SNAP household head' interchangeably to refer to this person.

and Ranney, 2000; Shapiro, 2005; Hamrick and Andrews, 2016; Beatty et al., 2019; Lee et al., 2022). This pattern where households exhaust their SNAP benefits shortly after receiving them is referred to as the *SNAP cycle*, and Shapiro (2005) shows that this cycle contributes to a 10 to 15 percent decline in calorie intake over the benefit month. Prior work of the SNAP cycle's effect on cognitive and health outcomes suggests that end-of-month food scarcity may increase the probability of injury through what I will refer to as the *resource availability* channel. Based on the terminal SSN digit I observe for each injured worker, I collapse the SNAP benefit month into approximately 5-day bins and construct a balanced daily panel of injury outcomes for cells of observably-similar workers.[4] I then measure changes in workplace injuries at different points in the benefit month relative to the first days after benefit receipt using a flexible set of dummy variables. The primary outcomes I consider are the probability of any injury and worker injury rates, the latter of which relies on annual employment estimates at the cell level.

I note that in order for an injury to enter the administrative record, the individual must both have been at work and gotten injured, and a worker's labor supply is potentially endogenous to the SNAP cycle. Marks et al. (2024) find that SNAP-eligible workers reduce hours and report more absences in the final days of the benefit month relative to the first week, which suggests that workers may actually reduce their exposure to occupational risk when their risk of injury is plausibly highest. Because I do not observe underlying labor supply, I attempt to disentangle these channels indirectly by comparing movement in the relative probability of severe versus less-severe injuries. Consider the simplest case where a worker is absent. Absence mechanically reduces exposure to all workplace risk, so the probability of injuries along the severity distribution should change proportionately in response. Similar logic can be applied if a worker decides to instead pick up an additional shift; increasing work exposure all else equal should increase severe and less-severe injuries proportionately. Therefore, any differential movement in the probability of injury along the severity distribution should reflect changes in resource availability.

Overall I am unable to detect a SNAP cycle effect on injury outcomes when pooling estimates across workers, and I find no consistent effect for workers between age 25 and 44. However, I estimate significant and economically meaningful effects for the youngest workers (18–24) and, less consistently, the oldest workers (45–54) in my sample, and the injury profiles of young workers differ sharply by gender. Women in this age group experience the most injuries in the first week of their benefit month, after which injury

[4] In my setting, 'observably-similar' refers to workers of the same gender, age range, and occupation who share a terminal SSN digit and have comparable wages. For example, one cell may contain all female waitresses between age 18 and 24 whose SSN ends in 4 and another cell contains all male roofers between age 35 and 44 whose SSN ends in 7.

rates fall over much of the month before trending back toward baseline in the final week. Injury rates for similar-aged men are flat until the middle of the benefit month and then rise: in the second half of the month, non-severe injuries are 17.6 percent and severe injuries 25.4 percent higher than in the two weeks after receipt. For both groups, severe and less-severe injuries move together in proportion to their baseline levels (I cannot reject equal proportional changes; $p = 0.28$ for men and $p = 0.68$ for women), which suggests that these profiles largely reflect gender differences in labor supply over the benefit month. Women aged 45 to 54 show roughly the mirror image of young women's profile, with injuries peaking in the third week of the benefit month, though these estimates are less precise.

However, the injury profile for men aged 45 to 54 indicates that food availability distinctly matters. On the day benefits are received, the rate of severe injury is 47 percent higher than in the following two weeks ($p = 0.002$), while the rate of less-severe injury is statistically unchanged, and the two effects are significantly different ($p = 0.003$). A change in hours worked would scale both types of injury proportionally, so this shift in the severity composition of injuries points to the resource availability channel.

It remains an open question why I do not find an effect for workers in the middle age groups. One possibility is that my eligibility metric may perform poorly at characterizing SNAP participation for this population of workers. If a significantly larger share of workers in these groups are not true SNAP recipients, then estimates for this sample would be disproportionately attenuated relative to others where workers are more accurately classified. Assuming this is not the case, another explanation could be that workers age 25 to 44 have significantly different household compositions than the youngest and oldest workers in my sample. They may be more often in two-adult households, which plausibly could reduce acute financial burden or allow households to adjust labor supply more flexibly when resources are scarce. They may also be more likely to have school-age children, and school meals should in theory slacken the household food budget. In an on-going analysis, I try to test this directly by comparing injury profiles for these workers during the summer and school year.

This work contributes primarily to two strands of literature. The first is the literature documenting reduced-form effects of the SNAP cycle on non-economic outcomes, and my paper is the first to my knowledge to link the timing of food assistance with safety outcomes. Much of the existing work in this area has found end-of-benefit month effects consistent with the resource availability mechanism (Gennetian et al., 2016; Carr and Packham, 2019; Cotti et al., 2020; Bond et al., 2022), though my findings are most consistent with those that find non-linear effects (Gassman-Pines and Bellows, 2018; Cotti et al., 2018). More

broadly, my findings also build upon work that studies how predictable non-work cash transfers affect labor supply decisions in the short-run. Marks et al. (2024) is the only other paper studying labor supply responses to the SNAP cycle, though there is also evidence of a labor supply response around receipt of social security payments (Beatty and Weill, 2024). My second contribution is to the body of research documenting the economic drivers of workplace injuries in the US. Several papers have examined the effect of minimum wage laws on occupational injuries, with mixed findings - both Davies et al. (2026) and Hradil (2018) find a positive relationship between the minimum wage and workplace injuries, while LaFiandra and Schwab (2023) finds a negative effect in the restaurant industry. Other papers have considered how economic conditions affect reporting behavior of workplace injuries (Card and McCall, 1996; Boone and Van Ours, 2006; Boone et al., 2011; Dillender, 2015). My work illustrates that the probability of workplace injury is responsive to short-run economic scarcity.

This paper proceeds as follows. Section 2 provides relevant background about SNAP and WC and how both programs operate in Oregon. In section 3, I discuss my data sources, provide summary statistics, and introduce my empirical strategy. Section 4 presents my main results and explores heterogeneity along several dimensions, and in section 5 I discuss my exploration of potential mechanisms. Finally, section 6 concludes and discusses future directions.

2 Background

2.1 Supplemental Nutrition Assistance Program

SNAP is funded through the Department of Agriculture's Food and Nutrition Administration (FNA), but the bulk of program administration is delegated to individual state agencies. For much of the program's history, benefits were issued as paper coupons in fixed dollar amounts that functioned as currency in exchange for eligible food items. Starting in the 1980s, states began to adopt the Electronic Benefit Transfer (EBT) system to deposit benefits directly to participants' bank accounts, and the 2008 Farm Bill established the EBT system as the only vehicle for issuing SNAP benefits.[5] Since then, benefits have been automatically deposited into participant accounts linked to debit-like EBT cards that can be used at authorized retailers. The changes implemented by the Farm Bill coupled with the onset of the Great Recession marked 2008 as a turning point in the program's history - participation grew by only 6 percent between fiscal years 2006 and

[5]Oregon was one of the early adopters of EBT and had implemented the system statewide by 1996.

2008 but by 43 percent between 2008 and 2010 (Jones, 2025).

SNAP is a means-tested program, similar to Medicaid and the Earned Income Tax Credit, so eligibility is determined by a household's financial means and family composition. Individual state agencies determine the specific eligibility criteria for participation in that state conditional on meeting federal requirements. Federal guidelines stipulates that households are eligible for SNAP if their gross monthly income is at or below 130 percent of the federal poverty line (FPL) and net income is below the FPL, but most states have set gross income requirements above the 130 percent requirement. SNAP participants must generally either be citizens or legal permanent residents, though states have some flexibility in theory to offer benefits to mixed-status households. In addition to income criteria, able-bodied adults without dependents (ABAWDs) must work 80 hours a month in order to receive benefits for longer than a 3 month period in 3 years. However, states were given the option to waive this requirement during the Great Recession and the recovery period if unemployment rates remained high.

State departments are also responsible for setting the monthly benefit disbursement schedule. The vast majority of states stagger disbursement throughout the month to minimize burden for retailers and reduce the probability of stock-outs, though a handful of less populated states distribute all benefits on a single day. Among the states that stagger disbursement, there is significant variation in both the timing and variable(s) used to randomly assign benefit days to participants. For example, Connecticut and Delaware both use the first letter of the client's last name to determine their benefit day, but Connecticut staggers benefit disbursement over the first three days of the calendar month while Delaware staggers issuance from the second through the 22nd. Alternatively, many states use characteristics of individuals' SNAP case numbers, social security numbers (SSNs), or birth dates to assign benefit days.

2.2 Workers' Compensation

Workers' compensation (WC) insurance is a state-regulated system that provides medical benefits and missed wages to employees who are injured at work or incur an illness attributable to work exposure. Importantly, injured workers are entitled to benefits regardless of fault or negligence. There is no federal law requiring employers to have WC insurance, but most state departments have laws mandating universal coverage, Texas being the exception (NASI, 2024).[6] Even though all employers are generally required to

[6]Using workers' compensation data from Texas, Cabral et al. (2022) estimate the demand for WC insurance and argue that mandating coverage may not be surplus-maximizing relative to elective coverage.

purchase WC insurance, there are typically exemptions for a subset of *employees*. State laws vary in which employees are exempted from coverage, though some of the common exemptions are private household workers, independent contractors, and casual workers. In 2022, the average cost to employers for providing WC insurance was 1 percent of covered wages, and the total cost to employers was approximately $103 billion.

Given the no-fault nature of WC benefits, an extensive body of economic research has attempted to size the degree of moral hazard in WC injury reporting. Krueger (1990b) and Krueger (1990a) show that increases in WC benefit generosity induces more participation and leads to workers' taking more days away from work after sustaining an injury, though Bronchetti and McInerney (2012) argue that the elasticity of participation to changes in benefit generosity is less economically meaningful than previously thought. Smith (1990) provides the first evidence of a 'Monday effect' where sprains and strains are more likely to be reported on Mondays, indicating that workers may be filing WC claims for off-the-job injuries, though Card and McCall (1996) find that uninsured workers are no more likely to report Monday injuries than those likely to have other medical coverage. However, more recently Dillender (2015) documents that reliance on WC for medical care does in part depend on availability of other health insurance coverage.

2.3 SNAP and WC in Oregon

Oregon was one of the early adopters of the EBT system (1996) and of broad-based categorical eligibility (BBCE) for SNAP participation (2000). Under BBCE, households that are eligible for Temporary Assistance for Needy Families (TANF) are automatically considered eligible for SNAP. Oregon households are eligible for SNAP if their gross monthly income is no higher than 185 percent of the FPL and their net monthly income is no higher than 100 percent of the FPL, regardless of household assets. Consequentially, Oregon has been consistently ranked in the top 5 states nationwide for SNAP accessibility (FNS, 2025). Almost 18 percent of the Oregon population participated in SNAP on average over my study period (see appendix figure A.2). At the state's peak in 2012, more than 1 in 5 persons received SNAP benefits and the total value of benefits issued exceeded $1.2 billion.

Oregon staggers SNAP benefit issuance over the first 9 days of the calendar month using the last digit of the primary account holder's SSN. Households for whom the account holder's SSN ends in 0 receive benefits on the first of the month, and those where the relevant digit d falls between 1 and 9 are issued benefits on calendar day d. Non-citizens without SSNs obtaining SNAP benefits on behalf of citizen children also

receive their benefits on the first of the month. In appendix figure A.1, I show how the SNAP benefit month overlaps with calendar time for a 31-day month.

All employers with at least one subject worker who is not an independent contractor are required to purchase WC insurance in Oregon. Workers are considered subject workers unless they meet one of the approximately 30 exempted conditions. Most of the exemptions pertain to sole proprietors, though privately employed household workers, volunteers for nonprofit or charitable organizations, and referees for recreational sports are also excluded from the subject worker definition. Employers may purchase coverage from one of the several hundred authorized insurers or they can opt to self-insure, conditional on being approved by Oregon's Department of Consumer and Business Services (DCBS). Premium rates in Oregon generally fall below the national median, which in 2024 meant that Oregon employers paid $0.89 per $100 of payroll while employers in states at the median paid a little over a dollar per $100 of payroll. However, employers and workers also each contribute roughly 1 cent per hour worked to Oregon's Workers Benefit Fund, which was established to supplement financial support for workers whose injuries meet certain extenuating circumstances.

When an injury occurs, the injured person fills out a first report of injury (FROI) form detailing the event and submits it to their employer. The employer then has 5 days to report the claim to the insurer *if* the injury required medical treatment beyond simple first aid. Employers are not required to report injuries that do not require any medical treatment. The insurer then has 60 days to accept or deny the claim and classify accepted claims as either disabling or non-disabling. Accepted disabling claims entitle workers to coverage for medical services and disability or death benefits, while workers with non-disabling claims are only entitled to coverage for medical services. After a decision has been made, the insurer must report all accepted disabling claims and all denied claims to Oregon's DCBS within 14 days. The insurer is not required to report accepted non-disabling claims to DCBS. The injured worker has 60 days to appeal decisions regarding the disability status of their claim or request a hearing to dispute a denial. Workers that are granted indemnity payments for lost wages receive installments equal to two-thirds of their weekly wage paid on a biweekly basis.

Indemnity payments compensate workers for lost wages after a disabling injury, and the exact award owed to the worker depends on whether a doctor has approved *any* work-related activity and on whether disability persists after the worker is deemed medically stationary. Most accepted disabling claims result in a *temporary* disability award, which entitles workers to payments for lost wages at two-thirds of their average

wage rate. If reduced or modified work is deemed permissible, workers are entitled to temporary *partial* disability (TPD) payments, but if all work-related activities are prohibited, the worker receives temporary total disability (TTD) payments. After the worker is medically stationary, any persistent disability that reduces capacity to work entitles the worker to *permanent* disability payments that are either partial (PPD) or total (PTD).

3 Methods

3.1 Data sources

My primary source of data are administrative records for all accepted disabling claims filed by workers in Oregon for injuries occurring between 2005 and 2018 that I have obtained from DCBS. For each claim, I observe the date the injury occurred, the date the employer was notified, the county where it occurred, the indemnity award, the industry of the employer, the cause of the injury (e.g., transportation accident vs. violent act), the nature of the injury (e.g., traumatic injury vs. infectious disease), and the part of body injured. I also observe information about each claimant, namely their gender, age (binned), occupation, average weekly wage, and the last digit of their SSN, which is the variable used to assign one's SNAP issuance day. If employers reported wages at a different interval, DCBS automatically converts the reported wage to the weekly equivalent, as the claimant's average weekly wage determines their indemnity payments. Occupations are reported as Standard Occupational Classification (SOC) titles which can vary over time, so I standardize all occupations to be consistent with the 2010 version published by the Bureau of Labor Statistics (BLS). Of the claims with non-missing occupation, over 90 percent of occupations could be linked to the most granular (6-digit SOC code) level, while the remaining were linked to 5-digit codes.

Ideally, I could observe whether or not each claimant was a SNAP participant when their injury occurred and if so, whether or not they were the policy holder for their household. The second-best case would be if I could observe their household size and gross income, which would allow me to directly impute SNAP eligibility. As none of this information is available to me, I instead use the variables I do observe to approximately identify SNAP eligible workers based on data from the American Community Survey (ACS), which are sourced from IPUMS (Ruggles et al., 2025). The ACS is a nationally representative survey conducted by the US Census Bureau since 2005, with an annual sample size of more than 3 million households, making it the most complete source of detailed microdata for US households. Among other

topics, ACS respondents are asked to report their wage and salary income, the number of weeks worked, and whether anyone in their household received SNAP benefits in the 12 months prior to enumeration. IPUMS also constructs a variable indicating where respondents' household income falls relative to the FPL. I restrict to ACS respondents in the 2005 to 2018 samples who were in the labor force, between the ages of 18 and 64, and who had worked for a wage or salary in the 12 months prior to enumeration. I calculate each respondents' "weekly wage" by dividing their reported annual wage and salary income by the number of weeks worked. While somewhat imprecise, this is actually a reasonably good approximation of the average weekly wage solicited by WC insurers. I then construct the weekly wage distribution in each year to assign respondents to their respective quintile.

I also rely on several secondary data sources to supplement this work. I use the FNA SNAP Quality Control (QC) database for fiscal years 2005 to 2018 to generate descriptive statistics of SNAP participants in Oregon over my study period (FNA, 2026). The SNAP QC data come from monthly random samples of active SNAP cases generated by each state department to be used in federal program audits. Each case record contains administrative information on the economic and demographic characteristics of all members of the SNAP household, including their age, gender, and employment status, and samples are constructed to be representative at the state level. The number of households sampled in each state-year depends on the average statewide monthly caseload for that fiscal year and whether or not the state was approved to submit a restricted sample. Oregon consistently had caseloads exceeding the maximum threshold, so I observe roughly 1,000 households every year. Since each record in the QC database is rigorously verified for federal use and samples are representative within-state, it can more reliably characterize SNAP households than the ACS or other household surveys.

Finally, I use BLS estimates on the number of injuries by days away from work (DAFW) for each nature of injury and injured body part to construct an injury severity index for claims in my sample. The reported estimates are based on data from the annual Survey of Occupational Injuries and Illnesses (SOII) from 2011 through 2018 (BLS, 2018a,b,c).[7] For each nature and body part classification, BLS reports the total number of injuries in each year and the number of injuries in incremental DAFW bins (1 day, 2 days, 3-5, 6-10, 11-20, 21-30, 31+). The BLS does not publish injury counts by nature $\times$ body part $\times$ DAFW, which would be the ideal metric to determine severity of my observed claims, so I instead use an iterative proportional fitting (IPF) algorithm to generate a fitted distribution that replicates national injury counts by nature $\times$ body

[7] 2011 is the earliest year these data are available.

part, nature × DAFW, and body part × DAFW (Deming and Stephan, 1940). I detail how I construct this index in appendix B.

3.2 Estimating SNAP eligibility

My data do not allow me to observe individual workers over time, so instead I use the ACS data to estimate the probability of being eligible for SNAP at the gender×age×wage×occupation cell level. I use age in 10-year bins, quintiles of real weekly wage (in 2010$), and SOC groups predominantly at the 5-digit, or broad group, level both to match what is observable in the claims data and to increase the sample size of cells. A subset of broad codes are aggregated in the ACS by the Census Bureau if there is either no exact match to a Census occupation code or if there were fewer than 10,000 workers nationally. To maintain comparability with ACS observations, I mirror this aggregation in my claims data. I ultimately estimate SNAP eligibility using 384 unique SOC groups; of these, 336 correspond exactly to broad SOC groups (5-digit) and 7 correspond exactly to minor SOC groups (3-digit).[8] The remaining groups are aggregations within minor groups that mirror ACS groups.[9]

For each cell, I nonparametrically estimate the share of workers that are in SNAP eligible households. For simplicity, I assume that households are SNAP eligible if they either reported SNAP receipt in the last year *or* if household income was at or below 185 percent of the FPL, which is the BBCE limit in Oregon. SNAP participation rates among the eligible working poor population in Oregon were consistently over 75 percent between 2005 and 2018, so though I am roughly estimating the probability of being *eligible* for SNAP, this is a reasonably close upper-bound on the probability of SNAP participation. When cells are comprised of many observations, this is an assumption-free and transparent approach to approximating the probability of SNAP eligibility for workers in that cell, but these estimates are increasingly noisy and unreliable as the number of observations-per-cell falls. To reduce the noise of estimates for sparse cells, I blend the empirical estimate with predictions obtained with a gradient boosting machine learning algorithm based on the same predictors, which I term the "GB estimate". Gradient boosting sequentially builds a set of individual decision trees, where each new tree is trained to correct the residuals of the trees before it. Each successive tree is fit to the gradient of a loss function, and predictions from all trees are then summed

[8]The 3-digit groups are post-secondary teachers (25-1000), other teachers and instructors (25-3000), other healthcare practitioners and technical occupations (29-9000), supervisors of personal care and service workers (39-1000), funeral service workers (39-4000), fishing and hunting workers (45-3000), and supervisors of transportation and material moving workers (53-1000). These SOC groups are rarely, if ever, reported at more granular levels in the ACS.

[9]For example, mathematicians (15-2020) and statisticians (15-2040) are aggregated to 15-20XX.

to produce the final output, with a learning rate scaling the contribution of each tree. Conversely, a random forest (RF) procedure builds many independent trees in parallel, each on a bootstrapped sample of the data, then averages their predictions. I found that GB performed better than RF at out-of-sample predictions of SNAP eligibility, which made it the preferred choice in my setting. The weight assigned to the empirical estimate when blending increases smoothly with the number of ACS observations for that cell (N). In particular, I construct the weights so that the empirical estimate and GB estimate are weighted equally when $N = 25$. The downside of this method is that I only generate eligibility probabilities for cells that exist in the ACS data.

Figure 1 compares the distribution of SNAP eligibility rates across the empirical (panel (a)), the GB (panel (b)), and the blended (panel (c)) specifications. The empirical distribution is relatively smooth, though it is clear that sparse cells generate excessive estimates at 0 and 1. The relative lumpiness of the GB distribution is an algorithmic feature. Like a random forest, the GB algorithm is an ensemble of regression trees so its predicted eligibility is a piecewise-constant function of the covariates and can therefore take only a finite menu of distinct values. The lumpiness in the distribution is more pronounced than a random forest would produce because boosting sums over multiple shallow, shrunken trees that are sequentially fit on a set of coarse predictors, whereas a random forest averages hundreds of deep, independent trees and thus yields a finer grid of output values. The bunching at end-points is resolved in the blended distribution, while the remainder of the distribution closely resembles that of the purely empirical estimates. This is evidence that the blending approach has the intended advantage of generating more plausible estimates for sparse cells.

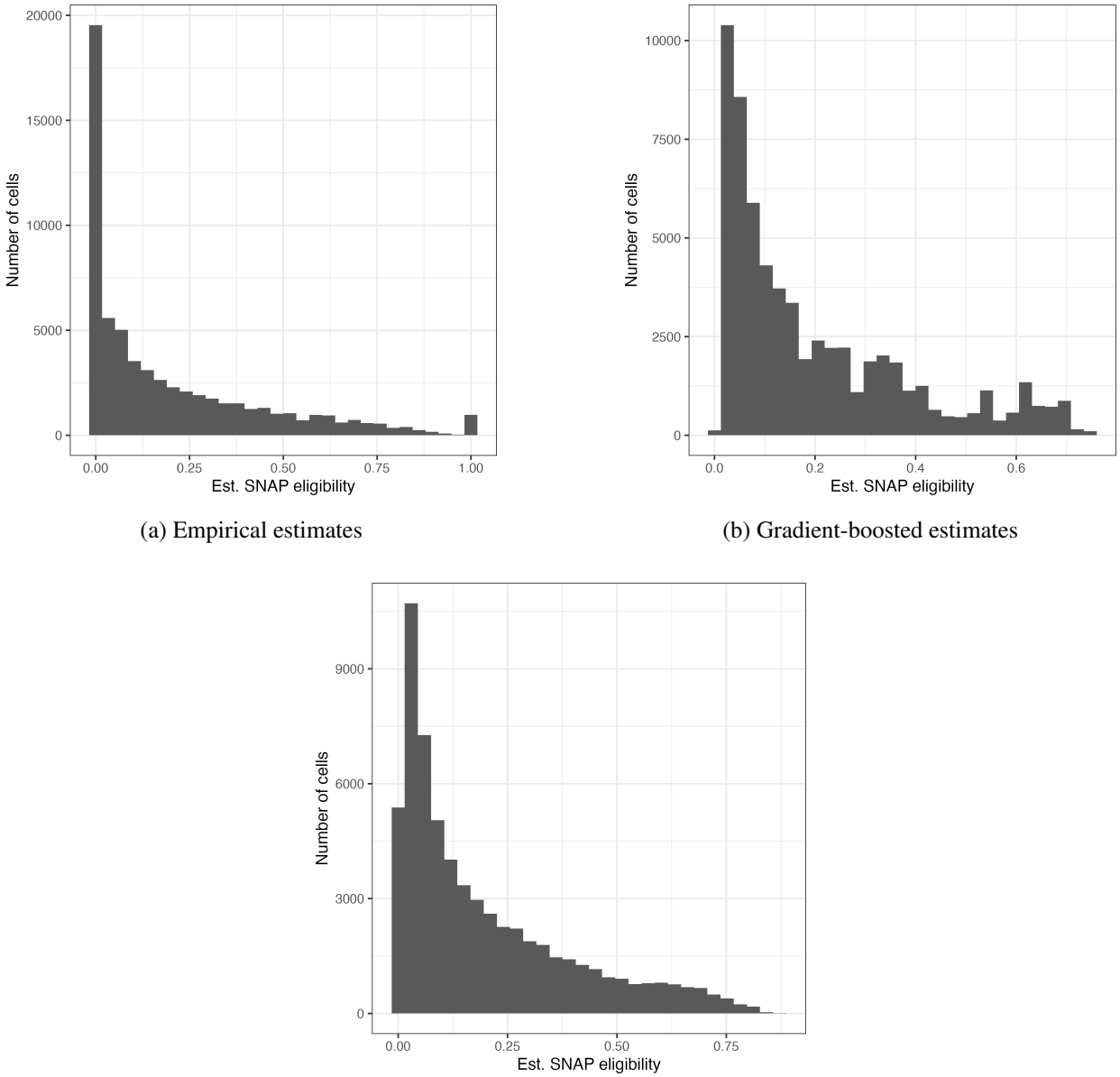
3.3 Analysis sample construction and summary statistics

I impose several limitations to arrive at the final set of claims in my analysis sample in order to facilitate my empirical strategy. I discuss the rationale behind each limitation below.[10]

I first restrict to claims filed by Oregon residents between the ages of 18 and 54, as over 90 percent of working female SNAP recipients and roughly 65 percent of working male SNAP recipients in this age range were their respective household head over this period.[11] Since SNAP households are assigned issuance days based on the SSN of the household head, I reduce potential measurement error in assigned benefit day by focusing on a population of workers who are more likely to be SNAP household heads. I rely on claimants'

[10] A data construction appendix is ongoing.

[11] Based on the author's analysis of data from the SNAP QC database, where a household head is considered 'working' if `WORKi` $= 1$ (available in FY 2015-2018) or if `EMPSTAi` > 2, `EMPSTBi` > 1, and earnings > 0 for FY 2005-2014.



(a) Empirical estimates

(b) Gradient-boosted estimates

(c) Blended estimates

Notes: Plots show the distribution of predicted probability of SNAP eligibility across observed ACS cells, where cells are unique combinations of gender, age bin, SOC group, and real weekly wage quintile. SNAP eligibility is defined at the worker-level and is equal to 1 if the worker's household reported SNAP participation *or* if their household income was at or below the BBCE eligibility threshold of 185 percent of the FPL. The empirical estimates are simple weighted shares of eligibility across all ACS respondents in that cell. The GB estimates are obtained from a gradient-boosting machine learning algorithm based on cell characteristics, and the blended estimates are weighted combinations of the empirical and GB predictions, where the relative weight assigned to the empirical estimates increases with size of the ACS sample.

Figure 1: Distribution of estimated $P(\text{SNAP eligibility})$ ($\hat{p}$) across ACS cells

observable characteristics - namely gender, age, occupation, and real weekly wage quintile - to determine their probability of SNAP eligibility, so I drop claims when occupation was unreported or could not be cleanly matched to a SOC group. A small share of the remaining claims have zero-valued wages, most of which are fatalities. I impute a subset of missing wage values when there were at least 5 other claims in that year for the relevant gender-age bin-occupation cell with non-missing wages, to which I assign the cell-year median, and I drop the remaining claims with zero-valued wages. Finally, I drop claims for workers whose corresponding gender-age-occupation-quintile cell had fewer than 100 weighted workers pooled across ACS samples.[12] After these restrictions, I am left with $n = 147,783$ (or 59 percent) of the 251,472 claims over this period.

The magnitudes of reduced-form effects of the SNAP cycle from prior work suggest that the effect size on occupational injuries will be small - recent studies find effects on test scores ranging from 0.01 to 0.17 of a standard deviation (Gassman-Pines and Bellows, 2018; Cotti et al., 2018; Bond et al., 2022), and effects on healthcare utilization between 1 and 2 percent (Arteaga et al., 2018; Cotti et al., 2020). In most of these cases, the authors were able to observe SNAP participation status and assigned issuance day for individuals in their respective samples. Given that I do not observe SNAP participation and effect sizes are expected to be small, I limit to claims where injuries were reported within 2 weeks ($n = 129,287$), and the claimants were likely eligible for SNAP. I use the worker-weighted median across all ACS cells ($\hat{p}_{50} = 0.22$) as the baseline threshold for inclusion in my sample, and I compare results under alternative thresholds. Oregon law requires workers to report injuries within 90 days of the event, though there are exceptional cases, so the former restriction serves to minimize potential measurement error that may arise from workers misremembering true injury dates. By focusing on workers who are more likely to be eligible for SNAP, my ITT estimates should more closely resemble the average treatment effect on the treated (ATT).

Table 1 provides summary statistics for claims filed by above-median workers. Between 89 and 96 percent of claims pertain to traumatic injuries that stem from a single incident. The relative share of non-traumatic claims, which include cumulative injuries and systemic illness, generally increases with age, as does injury severity and the share of claims earning permanent disability payments. The average probability of SNAP eligibility among workers above the population median ranges from 0.41 (45-54 women) to 0.52 (25-34 women). For women, average real weekly wages weakly increase with age from $331/week for workers age 18-24 to $392/week for workers age 45-54. Real wages similarly rise with age for men from

[12] The most common source of misalignment is in wage quintile, which may reflect reporting errors in the ACS or the FROIs.

	Women				Men			
	18–24	25–34	35–44	45–54	18–24	25–34	35–44	45–54
Share traumatic	0.96	0.94	0.90	0.89	0.95	0.93	0.90	0.89
	(0.21)	(0.24)	(0.3)	(0.31)	(0.22)	(0.26)	(0.3)	(0.32)
Severity index	0.50	0.51	0.52	0.53	0.51	0.52	0.53	0.54
	(0.08)	(0.08)	(0.08)	(0.09)	(0.10)	(0.10)	(0.10)	(0.10)
P(SNAP eligible)	0.49	0.52	0.48	0.41	0.43	0.44	0.44	0.48
	(0.09)	(0.15)	(0.14)	(0.12)	(0.1)	(0.15)	(0.15)	(0.13)
Real weekly wage (2010$)	331.05	377.4	384.43	391.6	414.27	491.65	506.27	421.44
	(143)	(148)	(155)	(152)	(196)	(194)	(164)	(153)
Share of claims by award								
Temporary disability	0.92	0.88	0.82	0.78	0.85	0.80	0.73	0.71
Permanent disability	0.05	0.08	0.13	0.17	0.11	0.16	0.20	0.22
Fatality	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01
Unknown	0.03	0.04	0.05	0.05	0.04	0.05	0.06	0.07
Share of claims by cause								
Bodily reaction and exertion	0.46	0.50	0.50	0.47	0.36	0.41	0.42	0.43
Contact with objects and equipment	0.19	0.15	0.14	0.13	0.34	0.30	0.26	0.23
Falls, slips, trips	0.18	0.20	0.23	0.28	0.16	0.17	0.20	0.22
Transportation accidents	0.03	0.03	0.03	0.02	0.05	0.04	0.05	0.04
All other causes	0.13	0.12	0.1	0.09	0.09	0.08	0.07	0.07
Share of claims by major SOC group								
Health support	0.25	0.28	0.22	0.23	0.02	0.02	0.02	0.03
Food preparation, service	0.22	0.19	0.18	0.17	0.12	0.09	0.07	0.09
Building cleaning, grounds maintenance	0.08	0.10	0.15	0.17	0.07	0.08	0.10	0.12
Retail sales	0.13	0.10	0.10	0.12	0.06	0.04	0.03	0.05
Farm, forestry, fishing	0.02	0.03	0.04	0.03	0.09	0.11	0.11	0.13
Construction	-	-	-	-	0.16	0.19	0.18	0.12
Transportation, material-moving	0.10	0.09	0.11	0.11	0.30	0.26	0.32	0.30
All other	0.20	0.21	0.20	0.18	0.17	0.21	0.16	0.16
Sample size								
N unique SOCs	63	69	63	59	66	80	59	47
N injuries	6,801	8,712	7,926	8,189	10,481	16,833	12,256	5,986

Notes: The baseline sample includes injuries by workers with above-median probability of SNAP-eligibility. Each cell reports the mean of the relevant statistic over the set of non-missing values, and corresponding standard deviations are shown in parentheses.

Table 1: Summary statistics: Baseline analysis sample

$414/week for workers age 18 to 24 to $506/week for workers between age 35 and 44, but the oldest men in my sample have relatively low earnings. The most common cause of injury is bodily reaction and exertion, which ranges from 36 to 50 percent of claims across groups. Young men have the highest incidence of injuries caused by contact with objects and equipment, while a mildly larger share of injuries for women stem from falls and other causes.

There is also predictable heterogeneity in the most common occupations by age and gender. Between 40 and 50 percent of women's injuries were for food service workers or healthcare support workers (e.g., nursing and home health aides), while injuries for men are similarly concentrated in transportation/material-

moving and construction occupations.[13] Farm, fishing, and forestry occupations account for approximately 10 percent of men's injuries, and 13 percent of injuries for men age 45 to 54.

Injuries in my sample are seasonal, peaking in summer months on average, and track the business cycle, falling during the Great Recession and rebounding as the economy recovered; the fixed effects described in section 3.4 absorb both patterns. I determine days since SNAP at the claimant level as the number of days between the date of injury and the most recent date of SNAP issuance, which I infer from the claimant's terminal SSN digit.

3.4 Empirical approach

This section describes my empirical approach to determine how injury risk changes over the SNAP benefit month for workers above the median of SNAP-eligibility probability. The SNAP cycle literature commonly uses days since last SNAP receipt to proxy for relative food scarcity, though the exact functional form can vary. A standard approach is to estimate a continuous effect of days since SNAP issuance with both linear and quadratic terms (e.g., Bond et al., 2022). Another common approach is to estimate differences across weeks of the benefit month using benefit week indicators, which avoids imposing a functional form on the relationship being studied. I follow a similar approach and compare injury outcomes at the start of the benefit month (days 0–5) to outcomes later in the month using 5-day bins $\mathcal{B}_k \equiv \{5k{+}1, 5k{+}2, 5k{+}3, 5k{+}4, 5k{+}5\}$ for $k = 1, ..., 5$.

Claims in my data are still largely anonymized and individual workers are not uniquely identifiable. To get a daily panel of injury outcomes, I instead aggregate injuries to the date level for groups of workers with key shared characteristics, namely their gender, age bin, occupation, and terminal SSN digit. The level of analysis is therefore the date (t) $\times$ occupation ($o \in O$) $\times$ demographic group ($d \in D$) $\times$ terminal SSN digit ($s \in [0, 9]$) level, where D is the set of unique gender $\times$ age combinations in my sample.[14] The basic regression specification is as follows:

$$
\begin{aligned}
Y_{s,d,o,t} &= \mathbf{X}'\boldsymbol{\beta} + \varepsilon_{s,d,o,t} \qquad (1)\\
&\equiv \sum_{k=1}^{5} \beta_k \mathbf{1}\{D_{s,t} \in \mathcal{B}_k\} + \delta'\mathbf{Z}_d + \gamma_{b(s)} + \gamma_{o,m(t)} + \boldsymbol{\gamma}_{\tau(t)} + \varepsilon_{s,d,o,t}
\end{aligned}
$$

[13] Material-moving is somewhat of an umbrella category, including occupations from warehouse workers to grocery store baggers.
[14] $D \in$ {female 18-24, male 18-24, female 25-34, male 25-34, female 35-44, male 35-44, female 45-54, male 45-54}.

where Y_{sdot} is a binary variable equal to 1 if cell sdo experienced an injury on date t ($1(\text{Injuries}_{s,d,o,t} > 0)$), an injury rate, or an injury count. $\mathbf{Z}_d$ is a vector of demographic indicators, namely gender, age, and the interaction. I include occupation $\times$ month-in-year ($\gamma_{o,m(t)}$) fixed effects to capture seasonal differences in injury risk across occupations, as well as a vector of other time fixed effects ($\boldsymbol{\gamma}_{\tau(t)}$) for year, day of month, day of week, and federal holidays. I also include issuance day fixed effects ($\gamma_{b(s)}$), though these are largely nuisance parameters. SSNs are randomly assigned so there is no obvious reason to expect level differences in injury outcomes between issuance day groups, but I include them to follow standard practice in the SNAP cycle literature (e.g., Carr and Packham, 2019; Bond et al., 2022). Standard errors are clustered at the issuance day $\times$ occupation level, so the number of clusters in the pooled sample is $G = 121 \cdot 9 = 1,089$. I estimate equation 1 on all workers pooled and separately by gender $\times$ age groups, in which case $\mathbf{Z}_d$ is dropped and the unit of analysis is s, o, t.

I estimate both weighted and unweighted regressions depending on the outcome of interest. To facilitate interpretation, my preferred estimates are obtained with unweighted ordinary least squares (OLS) when my outcome is binary and unweighted Poisson quasi-maximum likelihood (PQML) when I study injury rates. In the binary case, the β_ks can be interpreted as the percentage point effect on the probability of *any* injuries on a day in bin k relative to a day in the omitted period simply averaged across all s, d, o cells. The level difference between differently-sized cells is differenced out so there is not an econometric rationale to justify weights.

PQML is my preferred method to study the SNAP cycle effect on injury rates because coefficients are log changes in injury rates, which are easily converted to percentage changes. The dependent variable in this case is the count of injuries for cell s, d, o on date t, and the denominator, cell employment in year $y(t)$, is included on the right-hand side with its coefficient fixed to 1.[15]

With my regression equation in mind, I can now concretely discuss how measurement error in my dependent (injury rate) and independent (days since SNAP) variables will affect my estimates. The largest source of measurement error in my dependent variable arises from errors in annual employment based on ACS data, but unlike measurement error in independent variables, measurement error in an outcome variable is absorbed into the residual and reduces precision rather than generating bias. Critically, the identifying

[15] Since this outcome relies on annual employment estimates at the cell level, the sample underlying this specification is limited to gender-age-SOC cells that are observed in at least 10 (of 14) years of the ACS. I impute annual employment for missing years using a recency-weighted average of observed employment for above-median workers in that cell. I then allocate the worker population proportionally across terminal digit groups $s = \{0, \ldots, 9\}$.

variation in my design comes from days since benefit issuance, which varies at the st level, whereas the employment denominator varies only at the sdo-year level. Error in the annual employment estimate is therefore common across all days in a year and is orthogonal to the within-cell-year timing of benefit receipt that identifies my coefficients. Measurement error in annual employment would bias my estimates only if it were systematically correlated with the SNAP benefit month within a year, which is not the case by construction.

Two features of my data induce measurement error in treatment status, and both operate to make my estimates conservative. I discuss these sources at length and derive the implications for inference in appendix C. First, I do not observe SNAP participation. The analysis sample consists of workers whose observable characteristics place them above the median of predicted eligibility, so by construction it pools participants with non-participants, whose injury risk should not respond to the benefit cycle. Since a worker's terminal SSN digit is orthogonal to the determinants of eligibility and participation in SNAP, this pooling will attenuate the bin coefficients by the participation rate of retained workers, which I estimate to be roughly 0.40.[16] Measurement error will not change the shape of the estimated profile, so the estimates should be interpreted as lower bounds on effects for SNAP participants.

Second, Oregon assigns issuance days using the SNAP household head's SSN, so I will only accurately infer issuance days for injured workers who are the heads of their SNAP household. For other workers in the household, their true issuance day is a random draw from the schedule, and is independent from the worker's own terminal digit. Misallocation therefore contaminates every bin identically within a date and cancels in the within-benefit month contrasts that identify the bin effects. The degree of attenuation from this source of measurement error is expected to be heterogeneous across demographic groups due to differences in the probability of headship. Tabulations from the SNAP QC data suggest headship rates between 0.62 and 0.94 for workers across gender-age groups, with markedly lower values for men. Part of the cross-demographic variation in estimated magnitudes shown below therefore reflects differential misallocation rather than differential true effects. Combined, the two sources imply that the reported coefficients understate per-participant effects by a factor of roughly 2.7 to 4. However, I note that neither source affects the size of conventional hypothesis tests under the null but both reduce power in proportion to the attenuation factor. This means that my confidence intervals will be wide relative to studies with administrative participation

[16] This estimate comes from multiplying the mean eligibility probability among retained workers (≈ 0.45) by the average take-up rate estimated by FNA among eligible working households in Oregon (≈ 0.9 on average over my sample period, and above 0.75 in every year).

data.

4 Main results

Figure 2 plots the coefficients from estimating equation 1 on the sample of injuries by workers with above-median probability of SNAP-eligibility. Coefficients in the top panel were obtained from an unweighted linear probability model (LPM) on my binary outcome ($1(\text{Injuries}_{sdot}) > 0$), those in the middle panel correspond to employment-weighted least squares (WLS) estimates on injury rates, and coefficients in the bottom panel correspond to PQML estimates on injury *counts* but can be interpreted as percentage changes in injury *rates*. The remaining figures show only the LPM and PQML estimates, since the two injury-rate specifications yield similar profiles.

I find no effect of the SNAP cycle on injury outcomes when workers are pooled; and the point estimates are nearly all indistinguishable from zero. However, when I split the sample by demographic groups (figures 3a and 3b), I find that this null is in part explained by large and offsetting effects by gender and age. For the youngest group of female workers, injury outcomes fall over most of the SNAP benefit month before partially recovering in the final ten days. Injuries begin to meaningfully decline around the 11th day of the benefit month and start to trend back towards baseline in the last 10 days of the benefit month. Relative to the start of the month, the probability of injury increasingly falls to about 0.03 percentage points below baseline, which corresponds to an approximate 15 percent decline.[17] The injury-rate estimates show the same pattern, with injury rates about 13 percent below baseline at the trough. An inverse pattern emerges for the oldest group of female workers in my sample. The probability of injury steadily increases for these workers over the first two to three weeks of the benefit month and peaks in the 16 to 20 day period at nearly 0.02 percentage points above baseline (an 8 percent increase) before trending back down. Injury rates similarly peak during this window of the benefit month at roughly 9 percent over the baseline, though this effect is only marginally significant. I note that the rise in injuries relative to the first days of the benefit month suggests that workers in this subgroup actually work fewer hours in the reference period relative to the rest of the month, in which case the 0-5 bin is a non-representative baseline. Figure 3b shows that the age heterogeneity among male workers runs in the opposite direction: injuries for men aged 18 to 24 rise in

[17] I express LPM estimates as percent changes relative to the group's average number of injuries per cell-day, which approximates the mean of the binary outcome. PQML coefficients are log changes in injury rates; in the text I report them as percent changes, $100 \cdot (\exp(\hat{\beta}) - 1)$.

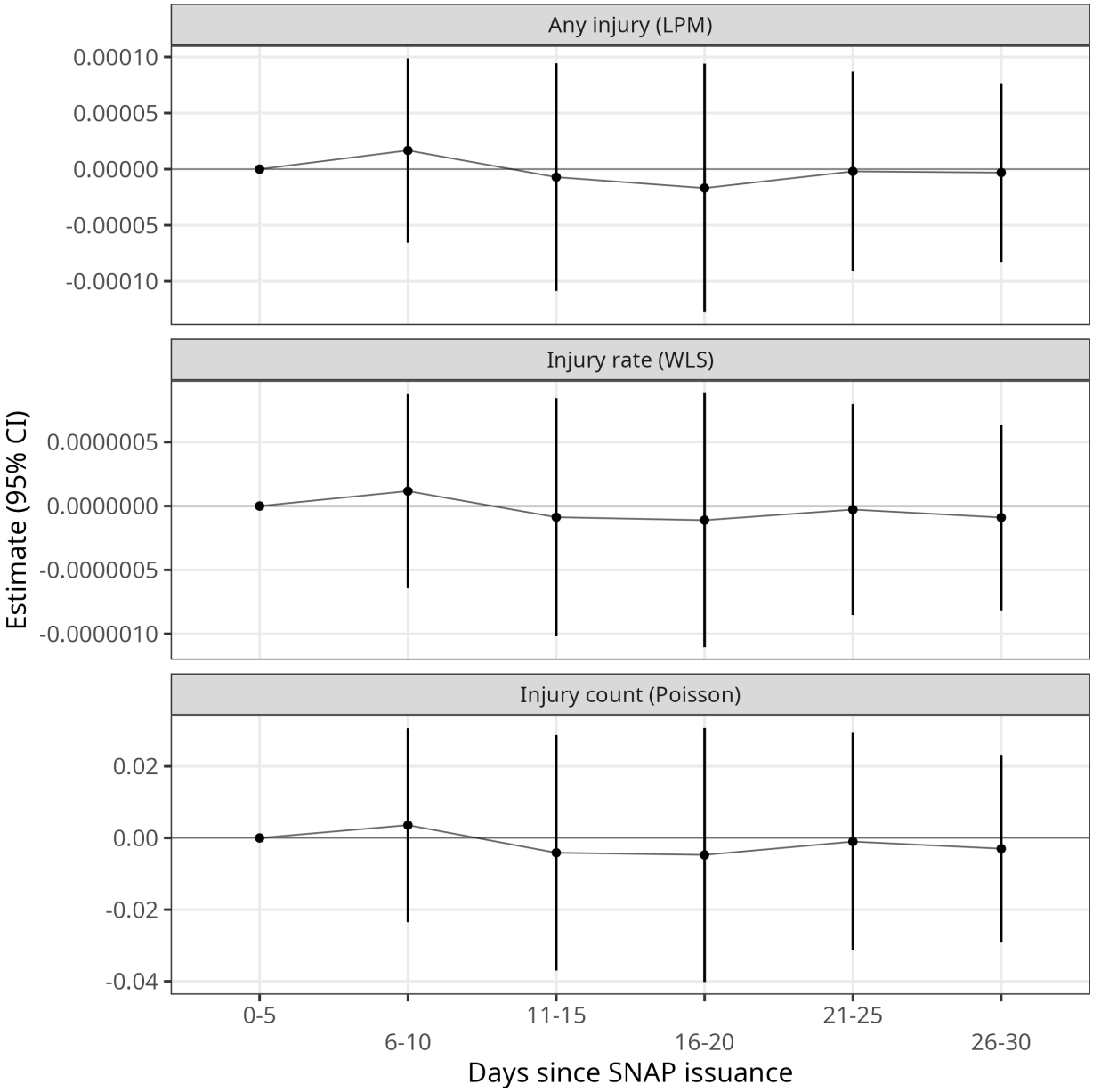


Figure 2: SNAP cycle effects on injury outcomes: Workers with $\hat{p} \geq 0.22$

Notes: Coefficients are obtained from estimating equation 1 on the pooled sample of workers above the population median probability of SNAP eligibility (see section 3.3 for a description of how eligibility probability was determined). Estimates in the top panel were obtained from an unweighted linear probability model (LPM), while the bottom two panels correspond to injury rate effects. Estimates in the middle panel come from employment-weighted least squares while those in the bottom panel were obtained from an unweighted PQML model. All regressions include issuance day, year, occupation$\times$month, day of month, day of week, and holiday fixed effects. Standard errors are clustered at the issuance day $\times$ occupation level, where $G = 1,089$.

the second half of the benefit month, while point estimates for men aged 45 to 54 are flat to negative, though both are less precise than the estimates for women.

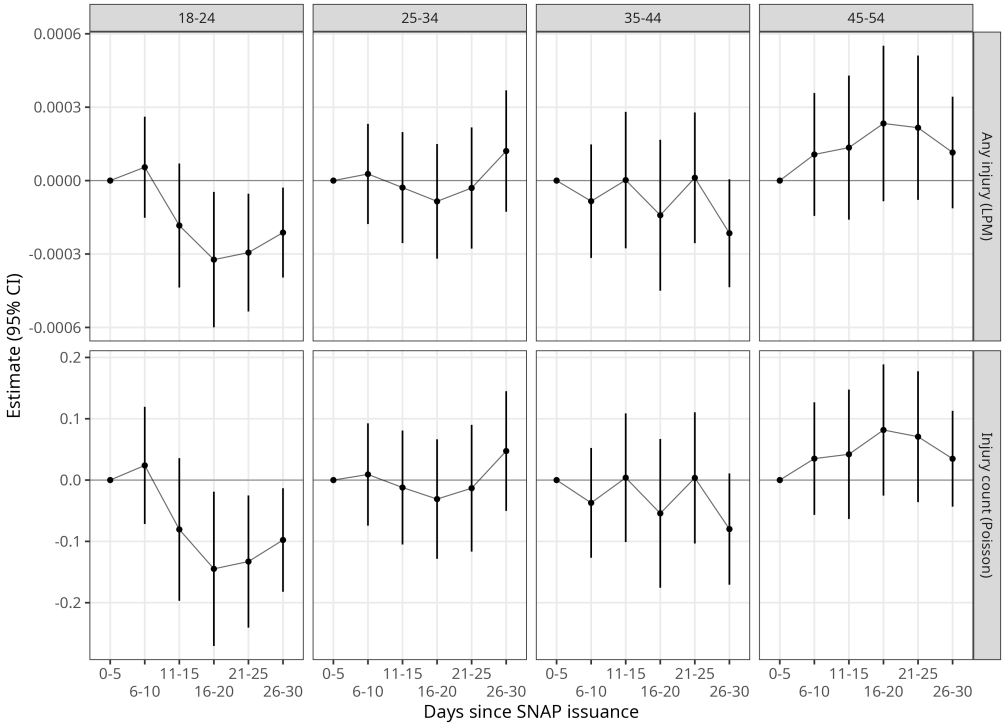
I expect that effects for men are relatively more attenuated because SNAP household-headship rates are uniformly higher for women, but this does not explain the wider confidence intervals. Figure 4 shows how effect sizes compare when the sample is restricted to workers above the 60th and 70th percentiles of the probability distribution relative to the median. Increasing the eligibility threshold increases the magnitudes of the injury profiles for the youngest men and the oldest women in my sample, which is consistent with the mean eligibility probability in samples rising from 0.43 (0.41) for workers above the median to 0.53 (0.51) for workers above the 70th percentile for men age 18-24 (women age 45-54). Though these magnitudes move as expected with higher expected rates of SNAP participation, I note that moving to higher thresholds mechanically changes the underlying sample of workers so the effect sizes should not be interpreted as directly comparable. Effects for all other workers are effectively unchanged at these higher thresholds, and I still find no definitive pattern for workers between 25 and 44.

4.1 Cell-specific thresholds

Since both gender and age are used as predictors to estimate SNAP eligibility, one may be concerned that the heterogeneity I find is driven by differences in true eligibility across subgroups. The gap between the population median and the group-specific median is largest for the youngest and oldest workers in my sample - the median eligibility for workers age 18 to 24 is approximately 0.5 while the median for workers age 45 to 54 is approximately 0.1. This results in nearly all young workers in my full sample meeting the population median threshold compared to only a quarter of older workers.

Figure 5 replicates the heterogeneity results from figure 3 using instead the relevant *group-specific* eligibility median as the sample inclusion threshold. Reassuringly, the shape of the benefit month profile is similar for all demographic groups regardless of threshold and in general, the magnitudes differ across samples in expected ways. Workplace injuries still fall over most of the benefit month for female workers between 18 and 24, and injury rates in days 21–25 are 22 percent lower than in the days immediately following benefit receipt. For men in this age group, injuries rise in the second half of the cycle, peak in the 16 to 20 days after benefit receipt, and stay persistently elevated above baseline through the benefit month. I still find mild evidence of an elevated injury trajectory for older female workers despite the change in the sampled population. In section 5, severe and non-severe injuries for this group both rise modestly in

(a) Women



(b) Men

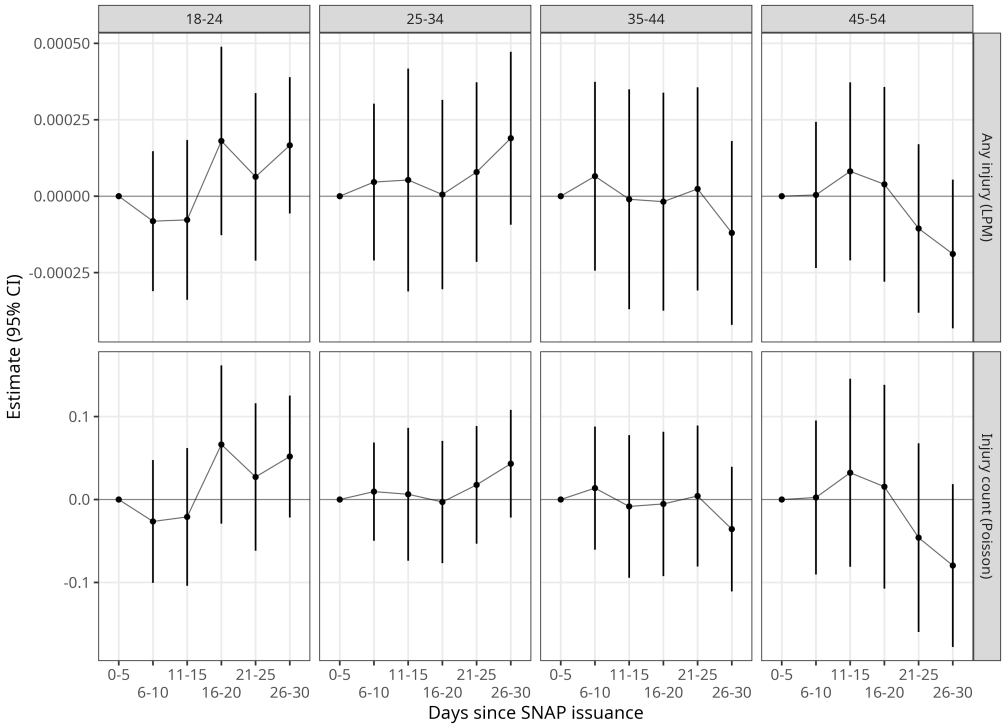
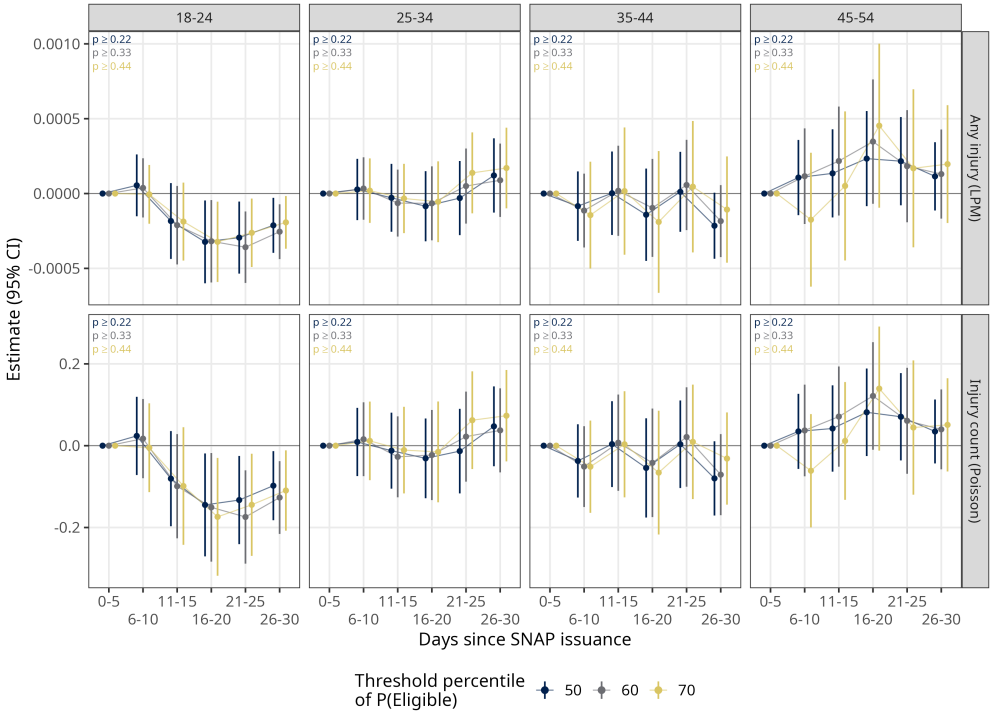


Figure 3: Heterogeneity by age and gender: Workers with $\hat{p} \geq 0.22$

Notes: Coefficients are obtained from estimating equation 1 on separate samples of workers with SNAP eligibility probability above the worker-weighted population median (see section 3.3 for a description of how eligibility probability was determined). The top panel shows estimates obtained from a linear probability model (LPM) and the bottom panel correspond to PQML estimates and are interpretable as log changes in injury rates. All regressions are unweighted and include issuance day, year, occupation×month, day of month, day of week, and holiday fixed effects. Standard errors are clustered at the issuance day × occupation level.

(a) Women



(b) Men

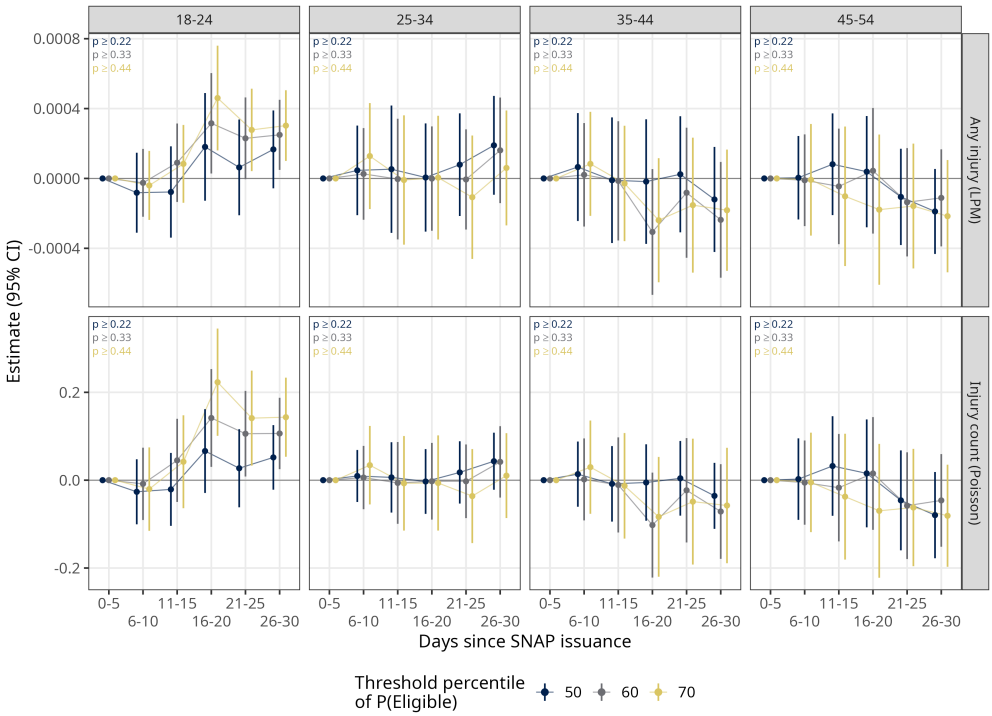
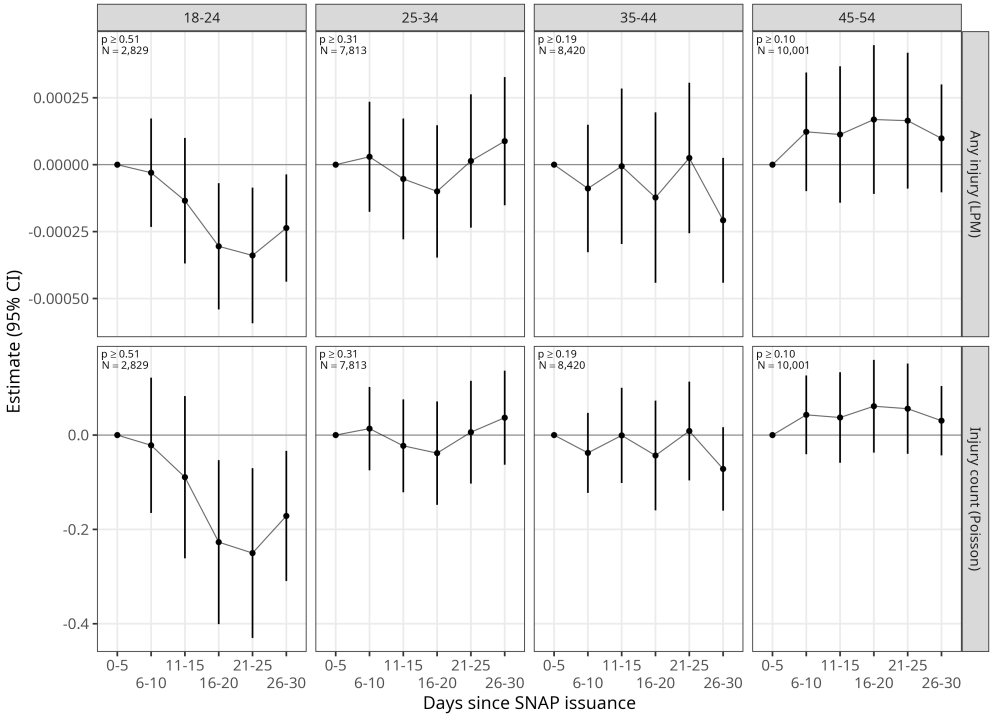


Figure 4: Heterogeneity by age and gender: Alternative sample thresholds

Notes: Coefficients are obtained from estimating equation 1 on separate samples of workers with SNAP eligibility probability above the specified percentile threshold, where 50 denotes the median (see section 3.3 for a description of how eligibility probability was determined). The top panel shows estimates obtained from a linear probability model (LPM) and the bottom panel correspond to PQML estimates and are interpretable as log changes in injury rates. All regressions are unweighted and include issuance day, year, occupation×month, day of month, day of week, and holiday fixed effects. Standard errors are clustered at the issuance day × occupation level.

(a) Women



(b) Men

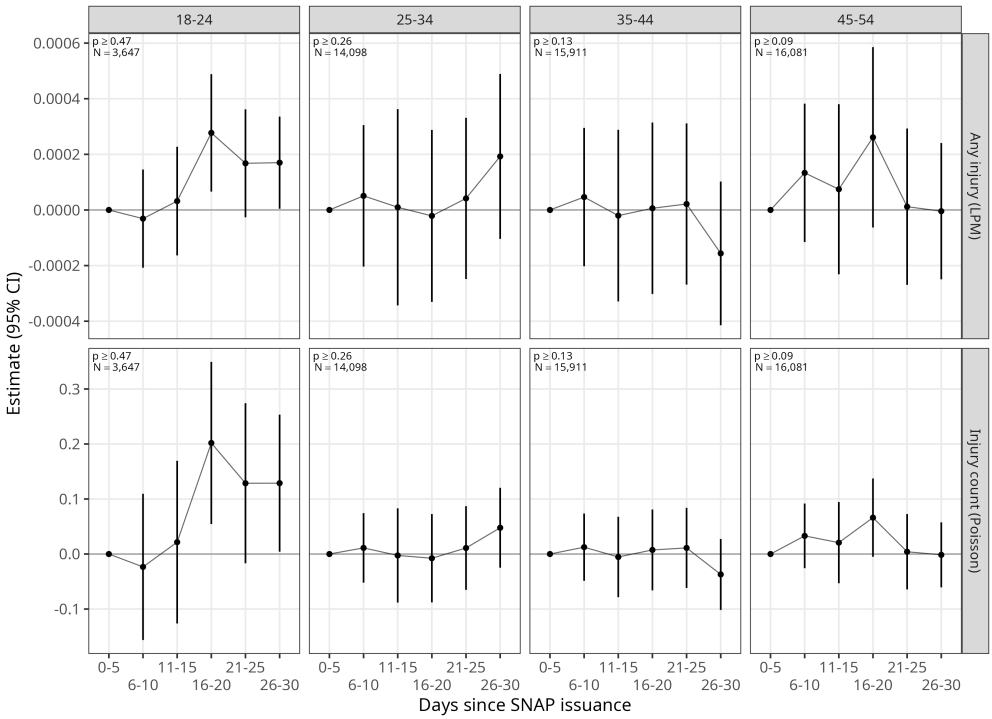


Figure 5: Heterogeneity by age and gender: Cell-specific medians

Notes: Coefficients are obtained from estimating equation 1 on separate samples of workers with SNAP eligibility probability above the worker-weighted median in each gender×age cell (see section 3.3 for a description of how eligibility probability was determined.) The top panel shows estimates obtained from a linear probability model (LPM) and the bottom panel correspond to PQML estimates and are interpretable as log changes in injury rates. All regressions are unweighted and include issuance day, year, occupation×month, day of month, day of week, and holiday fixed effects. Standard errors are clustered at the issuance day × occupation level.

the second half of the benefit month, but neither estimate is statistically significant, so I cannot distinguish between channels for these workers.

5 Mechanisms

This section aims to disentangle the resource availability channel from the labor supply channel by measuring the SNAP cycle impacts on the composition of injuries by severity. To motivate this with a thought experiment, suppose each worker faces an underlying distribution of injury severity conditional on an event. A reduction in hours worked reduces the probability of *any* injury but all else equal should leave the shape of the distribution unchanged. Likewise, a marginal increase in hours increases exposure to all workplace risk, which should rescale the severity distribution upwards in proportion to its baseline levels. Any change in the observed severity composition of injuries among workers in my sample in response to the SNAP cycle can therefore be attributable to changes in available resources. One could argue against this logic in an extreme case by noting that the marginal effect of increasing hours worked on injury outcomes is likely non-linear - a fatigued worker working extra overtime may actually face a higher risk of severe injury than a fresh worker who picks up an extra shift at the same resource levels. This is a reasonable objection, but I argue this is unlikely to matter in my setting because many of the workers in my sample are in service industry occupations where shift-work is standard. Shift schedules are often determined by the employer and generally are only flexible on the extensive margin. Moreover, recipient households are regularly re-evaluated to determine eligibility, so marginal workers may have incentive to keep their hours worked within certain bounds in order to meet work requirements (if applicable) and still remain eligible for SNAP.

I focus on traumatic injuries for this analysis, which comprise about 90 percent of claims. In light of the dilution shown in figure 4, I focus on workers whose probability of SNAP-eligibility is above the 70th percentile ($\hat{p} \geq 0.44$) to assess mechanisms. I automatically classify an injury as severe if it led to a fatality or if the claimant was awarded PTD benefits. For all other claims, I flag injuries as severe if the severity index value based on the nature and body part is in the top quartile of the observed distribution. The severity index is constructed to generate values between 0 and 1 with mean $= 0.50$, but the range of *observed* values is narrower $s \in (0.12, 0.81)$, with worker-weighted mean $= 0.53$ and 75th percentile $= 0.56$.[18] Severe injuries are by construction rare events, so in order to generate more reliable estimates when separating by injury

[18] See appendix B for how this index was constructed.

type, my preferred approach uses PQML and a coarser set of benefit month bins, where the day-of-receipt (i.e., days since SNAP $= 0$) and second half of the benefit month (days since SNAP > 15) is compared to the 1 to 15 days post-receipt. As I continue to estimate separate regressions by demographic group, for simplicity of notation I drop the d subscript and add in its place a subscript $v =$ Severe, Non-severe to denote injury type. I then estimate the following relative to the omitted period of the benefit month (days 1-15):

$$\text{Injuries}_{v,s,o,t} = \exp\Big(\beta_0^v \,\mathbf{1}(D_{s,t}=0) + \beta_H^v \,\mathbf{1}(D_{s,t}>15) + \log \text{Emp}_{s,o,y(t)} + \gamma_{b(s)} + \gamma_{v,o} + \gamma_{o,m(t)} + \boldsymbol{\gamma}_{\tau(t)}\Big)\,\varepsilon_{v,s,o,t} \quad (2)$$

where $v \in \{\text{severe}, \text{non-severe}\}$, the day-of-receipt ($\beta_0^v$) and second-half ($\beta_H^v$) effects are estimated separately for each injury type, and log cell employment enters as an offset. Relative to the PQML specification of equation 1, the only other change is the injury type $\times$ occupation fixed effects $\gamma_{v,o}$, which absorb level differences in severity across occupations. Figure 6 plots $100 \cdot (\exp(\hat{\beta}) - 1)$ for each type, with age on the vertical axis. To test whether injury composition changes, I use a Wald test of $\beta^{\text{severe}} = \beta^{\text{non-severe}}$ based on the clustered covariance matrix.

The day-of-receipt estimates shown on the left side of figure 6 are universally imprecise, particularly for severe injuries, where confidence intervals span roughly 50 percentage points or more. One exception stands out: for men age 45-54, rate of severe injury is 47 percent (CI = $[15.5, 87.2]$) higher on the day of benefit receipt relative to the omitted period, while non-severe injuries move in the opposite direction. A Wald test confirms that these coefficients are statistically distinct ($p = 0.0030$), which suggests that the day-of-receipt effect reflects a change in conditions on the job rather than in hours worked.[19] No other day-of-receipt estimate is distinguishable from zero and all are imprecise, so I consider the remaining results to be mostly uninformative rather than as evidence of no effect.

The estimates corresponding to the latter half of the benefit month are substantially more precise, and the youngest men in my sample continue to show the largest end-of-month effect. For these workers, the rate of non-severe injury is 17.6 percent higher [7.9, 28.1] at the end of the benefit month, and the rate of severe injury is 25.4 percent higher [10.9, 41.7]. I however fail to reject equality ($p = 0.28$), consistent

[19] I estimate 16 such differences (eight demographic groups and two periods). The day-of-receipt difference for men aged 45 to 54 remains significant under a Bonferroni correction for 16 tests (threshold $0.05/16 \approx 0.0031$).

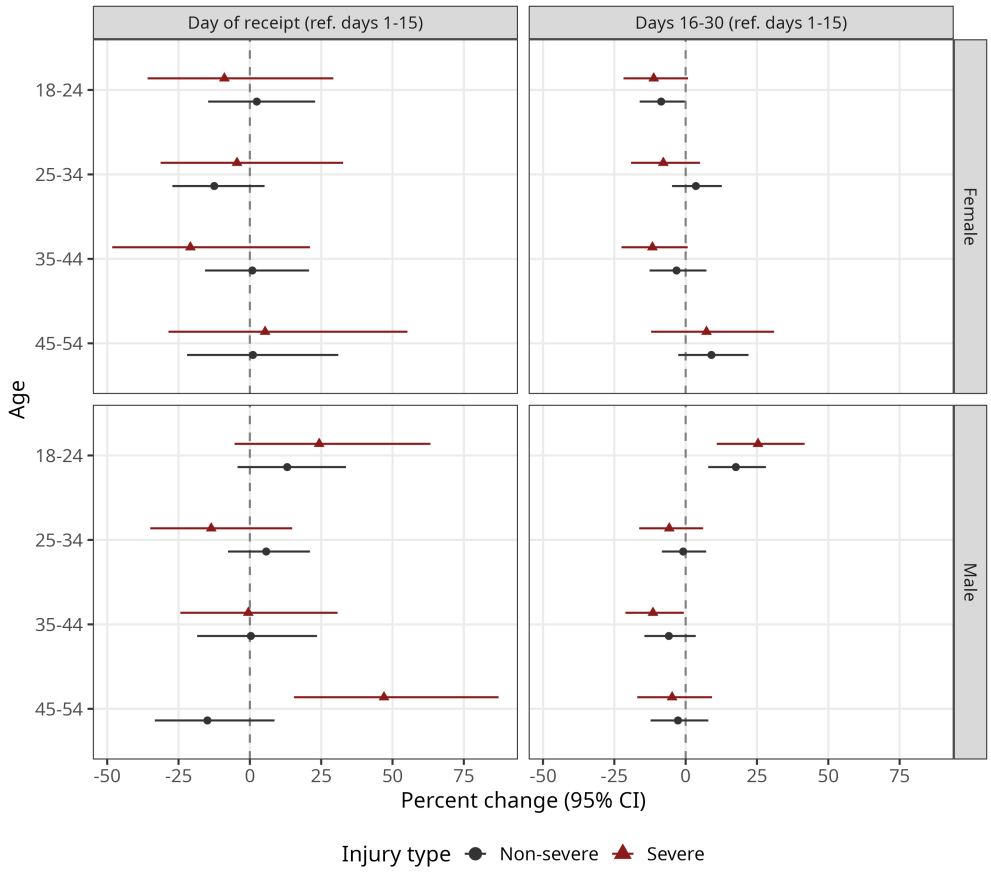


Figure 6: Heterogeneity by severity

Notes: Coefficients are obtained from estimating equation 2 by PQML on separate samples of traumatic injuries among workers with $\hat{p} \geq 0.44$ (70th percentile). Points show percent changes in injury rates, $100 \cdot (\exp(\hat{\beta}) - 1)$, on the day of receipt and in days 16–30 relative to days 1–15, separately for severe and non-severe injuries. All regressions include issuance day, year, occupation×month, occupation×injury type, day of month, day of week, and holiday fixed effects, with log cell employment as an offset. Standard errors are clustered at the issuance day × occupation level; the number of clusters ranges from 198 (45-54 women) to 513 (18-24 women).

with a proportional shift in risk rather than a change in injury composition. Women in this age group show the opposite sign - the rate of severe (non-severe) injury is 11.2 (8.6) percent lower on average at the end of the benefit month relative to the omitted period, and I similarly fail to reject that these effects are equal ($p = 0.68$). The only other coefficient significant at the 5 percent level suggests an 11.5 percent decline in the rate of severe injury for men age 35 to 44 relative to the two weeks following benefit receipt, which is puzzling but a result I do not emphasize given the number of comparisons and sample sizes.

Taken together, these results suggest that the benefit-cycle profiles of workers aged 18 to 24 largely reflect labor supply. For both young men and young women, severe and non-severe injuries move in the same direction and roughly in proportion, though I cannot rule out a smaller resource effect layered on

top. The clearest evidence for a resource availability channel comes from men aged 45 to 54, for whom the severity composition of injuries shifts on the day benefits are received. The point estimate for non-severe injuries on that day is negative, so if anything these workers reduce their exposure, and the rise in severe injuries is unlikely to reflect additional hours. Although benefits are typically deposited at midnight, a worker may have little opportunity to shop or eat a full meal before a shift, so the day of receipt may effectively be the last day of the previous month's scarcity. An alternative explanation is that benefit receipt itself changes behavior in ways that raise risk, which I cannot distinguish with the current data.

6 Discussion

This paper asks whether food availability affects the incidence and severity of workplace injuries for low-wage workers. Using Oregon's staggered SNAP issuance schedule and administrative WC claims, I compare injury outcomes for otherwise similar workers assigned to different points in their benefit month. Because I cannot observe participation directly, my estimates are ITT effects for workers with above-median predicted eligibility and likely understate effects for participants.

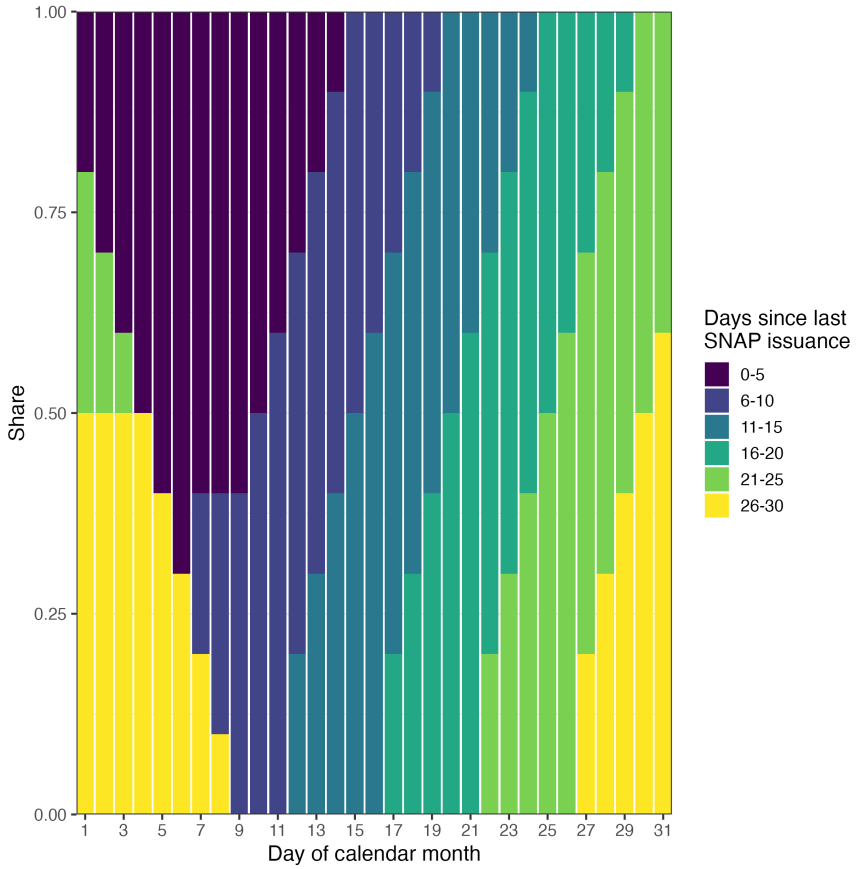
Workplace injury data come from administrative WC claims records, which do not contain sufficient information to identify SNAP participants directly. I instead limit my sample to workers whose observable characteristics place them in the top half of the probability distribution, which means that my estimates have an ITT interpretation and likely constitute a lower bound for the true effect for SNAP-participating workers. When predicted-eligible workers are pooled, I find no evidence that injury risk varies over the benefit month. However, this null is in part attributable to heterogeneity by gender for the youngest workers and oldest workers in my sample. For women aged 18 to 24, injury rates fall over most of the benefit month. This decline is largest for women whose probability of SNAP eligibility exceeds 0.5, where injuries fall by 20 to 22 percent below baseline by the third week of the benefit cycle before trending back upwards. I find the opposite pattern for men in the same age group with comparable eligibility probability: injury rates are relatively flat over the first two weeks of the benefit month and then rise sharply and remain elevated through the second half of the benefit month. Both patterns are robust to alternative sample thresholds, and the profile for young men grows larger as expected SNAP participation within the underlying sample rises. The severity analysis suggests that both profiles operate primarily through labor supply: severe and non-severe injuries rise together for young men, by 25.4 and 17.6 percent in the second half of the benefit

month, and fall together for young women. Young women may face time constraints, such as coursework or child care, that make it optimal to front-load their hours when resources are plentiful; distinguishing this explanation from others is a goal of ongoing work.

The clearest evidence that food availability itself matters comes from men aged 45 to 54. On the day benefits are received, their rate of severe injury rises by 47 percent relative to the following two weeks while non-severe injuries do not, a shift in the severity composition of injuries that a change in hours worked alone would not produce. The probability of injury also rises over the benefit month for women in this age group, but the severity estimates are too imprecise to separate channels for these workers. I find no consistent injury cycle for workers aged 25 to 44. These workers are more likely to have dependents, so they may be more insulated from work requirements, and school meals may relax their household food budgets; ongoing work tests whether injury patterns for this group differ between the school year and summer.

The next step to advance this work is to test which policy levers are most effective at mitigating injury risk for hungry workers. The American Recovery and Reinvestment Act was an economic stimulus package enacted in 2009 and in addition to increasing SNAP benefit amounts, it allowed states to more easily waive ABAWD work requirements. Future work will leverage the temporal and geographic variation in exposure to Oregon's ABAWD time limits during and after the Great Recession to link my injury results directly to differential exposure to work requirements across groups. I have also obtained WC claims data from Idaho, a state which changed its SNAP disbursement schedule twice over this period - in 2009, the state transitioned from distributing benefits over the first five days of the month to a single-day disbursement design, and then switched to staggering disbursal over *ten* days in 2016. This offers a complementary empirical design that would allow me to study two distinct shocks to benefit timing on workplace safety outcomes.

Taken together, my results suggest that the benefit cycle affects workplace safety, though more work is needed to determine the optimal policy response. Workplace injuries are costly for employers and especially for workers and their families, so if cyclicality in food availability affects the probability that a worker is injured, then the issuance schedule has consequences that neither the safety nor the SNAP literature has considered. These effects appear concentrated among workers aged 18 to 24 and 45 to 54. Recent federal legislation has extended SNAP work requirements to older adults, so workers in the latter group are increasingly likely to face them, making it more important to understand how food assistance shapes their safety on the job.



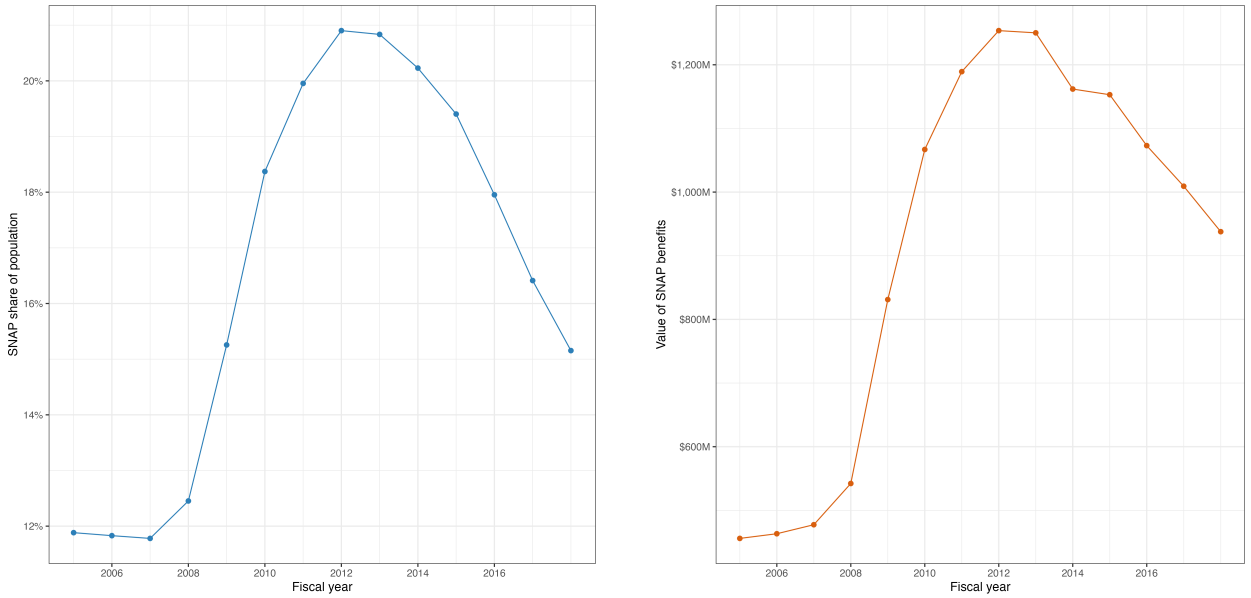
Notes: Oregon staggers distribution of SNAP benefits over the first nine days of the calendar month based on the terminal digit of the primary account holder's SSN. Terminal digits that end in 1 to 9 map directly to assigned issuance days, while households whose relevant terminal digit is 0 also receive benefits on the first of the month.

Figure A.1: Overlap between 31-day calendar month and SNAP benefit month in Oregon

A Additional Tables and Figures

B Constructing the severity index

I measure the severity of each nature $\times$ body part type by how long injuries of that type keep workers away from work, expressed as a rank. Data on injury counts by nature of injury and body part come from the BLS SOII (Table R19), and data on days away from work (DAFW) by nature of injury and by body part injured come Tables R67 and R68 for 2011 through 2018. The latter tables contain for each injury nature and body part, the number of *national* cases with DAFW across seven duration bins: 1, 2, 3–5, 6–10, 11–20, 21–30, and 31 or more days. A severity index based on the expected DAFW for the injury nature or body part alone would be incomplete; finger injuries generally result in fewer days off because fingers mostly get cuts and



(a) SNAP participation rate

(b) Total value of SNAP benefits

Notes: Data on SNAP participation and benefit amounts come from FNA's SNAP Monthly State Participation and Benefits Summary, while overall population estimates come from FRED.

Figure A.2: SNAP in Oregon: 2005–2018

bruises, and fractions alone predict more days off because this category includes major bone fractures. Since a finger fracture is more severe than a finger cut but less severe than a leg fracture, the ideal severity index should account for this nuance. The complication is that BLS does not publish DAFW statistics by nature and body part jointly. I instead estimate the joint distribution of nature $\times$ body part across DAFW bins using an iterative proportional fitting (IPF) procedure (Deming and Stephan, 1940), which uses maximum-likelihood estimation to iteratively generate weights for each two-way combination in order to replicate the observed two-way counts. In practice, this procedure requires the presently untestable assumption that moving an injury from one body part to another changes the odds of a longer work absence by the same factor for every nature. This assumption is more plausible for some natures (e.g., cuts and bruises) than for others; the difference in severity between an amputated finger and an amputated leg is arguably larger than the difference between a bruised finger and bruised leg. However, given that I currently use this index to categorize injuries rather than as an outcome variable, I view this as a second-order concern.

With the fitted distribution for each nature $\times$ body-part combination across duration bins, each type is then ranked relative to the national distribution for all injuries. Let w_{jb} be the share of type j's cases in bin b, and F_b the cumulative share of all national DAFW cases in bins 1 through b. Each bin is assigned

	Women			Men		
	N	% of Gender	% of Gender & Age	N	% of Gender	% of Gender & Age
Total workers	881,390	1.00		667,739	1.00	
Head of HH	802,847	0.91		452,499	0.68	
Under 25	153,255	0.17	0.85	69,514	0.10	0.63
25-34	266,832	0.30	0.94	129,335	0.19	0.64
35-44	183,990	0.21	0.92	93,908	0.14	0.62
45-54	123,463	0.14	0.92	88,442	0.13	0.77
55-64	63,481	0.07	0.90	62,923	0.09	0.81
65+	11,826	0.01	0.89	8,377	0.01	0.88
Not head of HH	78,543	0.09		215,240	0.32	
Under 25	26,501	0.03	0.15	41,506	0.06	0.37
25-34	16,591	0.02	0.06	72,729	0.11	0.36
35-44	16,462	0.02	0.08	58,763	0.09	0.38
45-54	10,560	0.01	0.08	26,446	0.04	0.23
55-64	6,988	0.01	0.10	14,665	0.02	0.19
65+	1,441	0.00	0.11	1,131	0.00	0.12

Table A.1: Tabulation of workers in SNAP participating households, Oregon 2005-2018

Note: This table shows total weighted counts of male and female workers participating in SNAP by age and household-head status.

Source: Author's own analysis of SNAP Quality Control data from FNA for fiscal years 2005-2018.

its mid-rank percentile in the national distribution, $\bar{p}_b = (F_{b-1} + F_b)/2$, and define type j's severity as the average percentile of its cases:

$$S_j = \sum_b w_{jb}, \bar{p}_b.$$

The bin scores $\bar{p}_b$ are *ridits*, which were named to connote similarity with *probits* and *logits*, and this concept was first introduced by Bross (1958) to construct ordinal ranks for variables "which can neither be optimally described by a nominal classification nor can be subjected to refined to refined measurement systems" (Kantor et al., 1968). S_j is the mean ridit of type j against the national reference distribution. It therefore has a direct probability interpretation. This construction has four advantages over simpler summaries. First, it uses only the ordering of the bins, so it needs no assumed value for the open-ended 31-or-more-days bin. That bin holds 29 percent of national cases, and any mean-days measure would depend on how it was top-coded. Second, it uses each type's full duration distribution rather than its median. Medians cannot separate the most serious injuries, because fractures, meniscus tears, and herniated discs all have medians inside

TABLE A.2—SNAP CYCLE EFFECTS ON INJURY OUTCOMES: WORKERS WITH $\hat{p} \geq 0.22$

	Any injury (pp) (1)	Injury rate (per 100,000) (2)	Injury count (3)
	OLS		Poisson
Days 6–10	0.0017	0.0116	0.0036
	(0.0042)	(0.0387)	(0.0138)
Days 11–15	−0.0007	−0.0087	−0.0041
	(0.0052)	(0.0475)	(0.0168)
Days 16–20	−0.0017	−0.0110	−0.0047
	(0.0056)	(0.0507)	(0.0181)
Days 21–25	−0.0002	−0.0027	−0.0010
	(0.0045)	(0.0421)	(0.0155)
Days 26–30	−0.0003	−0.0090	−0.0030
	(0.0041)	(0.0370)	(0.0134)
Fixed effects			
Issuance day	Yes	Yes	Yes
Year	Yes	Yes	Yes
Occupation × Month	Yes	Yes	Yes
Day of month	Yes	Yes	Yes
Day of week	Yes	Yes	Yes
Holiday	Yes	Yes	Yes
Gender × age indicators	Yes	Yes	Yes
Observations	25,207,090	25,207,090	24,358,210
Injuries	76,928	76,928	76,928
Clusters	1,089	1,089	1,089
R^2	0.008	0.002	
Pseudo R^2			0.144

Notes: Coefficients are obtained from estimating equation 1 on the pooled sample of workers with $\hat{p} \geq 0.22$, relative to days 0–5 of the benefit month. Column (1) is an unweighted linear probability model for any injury (in percentage points); column (2) is employment-weighted least squares on the injury rate (per 100,000 worker-days); column (3) is PQML on injury counts with log cell employment as an offset, interpretable as log changes in injury rates. All regressions include issuance day, year, occupation × month, day of month, day of week, and holiday fixed effects. Standard errors clustered by issuance day × occupation in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

TABLE A.3—HETEROGENEITY BY AGE AND GENDER: WORKERS WITH $\hat{p} \geq 0.22$ (LPM)

	Any injury (pp)							
	18–24		25–34		35–44		45–54	
	Female (1)	Male (2)	Female (3)	Male (4)	Female (5)	Male (6)	Female (7)	Male (8)
Days 6–10	0.0055	−0.0082	0.0027	0.0046	−0.0084	0.0065	0.0107	0.0004
	(0.0105)	(0.0117)	(0.0104)	(0.0131)	(0.0118)	(0.0157)	(0.0128)	(0.0122)
Days 11–15	−0.0184	−0.0078	−0.0029	0.0053	0.0002	−0.0010	0.0135	0.0081
	(0.0129)	(0.0133)	(0.0115)	(0.0186)	(0.0142)	(0.0183)	(0.0150)	(0.0148)
Days 16–20	−0.0323**	0.0181	−0.0085	0.0005	−0.0142	−0.0018	0.0233	0.0039
	(0.0141)	(0.0157)	(0.0119)	(0.0158)	(0.0157)	(0.0181)	(0.0162)	(0.0162)
Days 21–25	−0.0294**	0.0063	−0.0030	0.0079	0.0011	0.0024	0.0216	−0.0106
	(0.0123)	(0.0140)	(0.0126)	(0.0150)	(0.0136)	(0.0169)	(0.0150)	(0.0140)
Days 26–30	−0.0212**	0.0166	0.0121	0.0190	−0.0215*	−0.0120	0.0115	−0.0189
	(0.0094)	(0.0114)	(0.0126)	(0.0144)	(0.0112)	(0.0153)	(0.0116)	(0.0124)
Fixed effects								
Issuance day	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Occupation × Month	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day of month	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day of week	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Holiday	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,170,060	3,374,580	3,476,840	3,988,140	3,067,800	2,914,410	2,812,150	2,403,110
Injuries	6,792	10,481	8,703	16,787	7,823	12,215	8,141	5,986
Clusters	558	594	612	702	540	513	495	423
R^2	0.009	0.014	0.014	0.018	0.011	0.014	0.012	0.006

Notes: Coefficients are obtained from estimating equation 1 separately for each gender × age group, relative to days 0–5 of the benefit month, and are linear probability model estimates for any injury, in percentage points. Samples include workers with $\hat{p} \geq 0.22$, the worker-weighted population median. All regressions include issuance day, year, occupation × month, day of month, day of week, and holiday fixed effects. Standard errors clustered by issuance day × occupation in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

TABLE A.4—HETEROGENEITY BY AGE AND GENDER: WORKERS WITH $\hat{p} \geq 0.22$ (PQML)

	Injury count							
	18–24		25–34		35–44		45–54	
	Female (1)	Male (2)	Female (3)	Male (4)	Female (5)	Male (6)	Female (7)	Male (8)
Days 6–10	0.0239	−0.0265	0.0091	0.0095	−0.0372	0.0137	0.0350	0.0024
	(0.0488)	(0.0378)	(0.0425)	(0.0302)	(0.0457)	(0.0379)	(0.0468)	(0.0473)
Days 11–15	−0.0806	−0.0210	−0.0122	0.0063	0.0038	−0.0084	0.0420	0.0323
	(0.0593)	(0.0423)	(0.0474)	(0.0409)	(0.0535)	(0.0439)	(0.0538)	(0.0578)
Days 16–20	−0.1448**	0.0663	−0.0310	−0.0030	−0.0543	−0.0053	0.0817	0.0155
	(0.0642)	(0.0487)	(0.0497)	(0.0376)	(0.0619)	(0.0444)	(0.0546)	(0.0627)
Days 21–25	−0.1329**	0.0272	−0.0133	0.0177	0.0035	0.0042	0.0707	−0.0460
	(0.0550)	(0.0454)	(0.0527)	(0.0362)	(0.0546)	(0.0434)	(0.0544)	(0.0580)
Days 26–30	−0.0977**	0.0518	0.0474	0.0432	−0.0799*	−0.0357	0.0348	−0.0796
	(0.0431)	(0.0375)	(0.0498)	(0.0331)	(0.0463)	(0.0383)	(0.0399)	(0.0501)
Fixed effects								
Issuance day	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Occupation × Month	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day of month	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day of week	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Holiday	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,637,590	2,711,660	2,762,970	3,414,330	2,535,470	2,612,860	2,318,380	2,104,760
Injuries	6,792	10,481	8,703	16,787	7,823	12,215	8,141	5,986
Clusters	558	594	612	702	540	513	495	423
Pseudo R^2	0.120	0.135	0.138	0.146	0.123	0.140	0.134	0.091

Notes: Coefficients are obtained from estimating equation 1 separately for each gender × age group, relative to days 0–5 of the benefit month, and are PQML estimates on injury counts with log cell employment as an offset, interpretable as log changes in injury rates. Samples include workers with $\hat{p} \geq 0.22$, the worker-weighted population median. All regressions include issuance day, year, occupation × month, day of month, day of week, and holiday fixed effects. Standard errors clustered by issuance day × occupation in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

TABLE A.5—HETEROGENEITY BY AGE AND GENDER: CELL-SPECIFIC MEDIANS (LPM)

	Any injury (pp)							
	18–24		25–34		35–44		45–54	
	Female (1)	Male (2)	Female (3)	Male (4)	Female (5)	Male (6)	Female (7)	Male (8)
Days 6–10	−0.0030	−0.0031	0.0030	0.0051	−0.0089	0.0046	0.0123	0.0133
	(0.0103)	(0.0090)	(0.0105)	(0.0129)	(0.0121)	(0.0127)	(0.0113)	(0.0127)
Days 11–15	−0.0135	0.0032	−0.0053	0.0010	−0.0006	−0.0020	0.0113	0.0075
	(0.0119)	(0.0099)	(0.0115)	(0.0180)	(0.0148)	(0.0157)	(0.0130)	(0.0156)
Days 16–20	−0.0305**	0.0277**	−0.0100	−0.0021	−0.0123	0.0006	0.0169	0.0261
	(0.0120)	(0.0107)	(0.0126)	(0.0157)	(0.0162)	(0.0157)	(0.0141)	(0.0165)
Days 21–25	−0.0339***	0.0168*	0.0014	0.0042	0.0025	0.0021	0.0164	0.0012
	(0.0129)	(0.0099)	(0.0127)	(0.0148)	(0.0143)	(0.0148)	(0.0129)	(0.0143)
Days 26–30	−0.0237**	0.0170**	0.0088	0.0193	−0.0208*	−0.0156	0.0098	−0.0004
	(0.0102)	(0.0084)	(0.0122)	(0.0151)	(0.0119)	(0.0132)	(0.0103)	(0.0125)
Fixed effects								
Issuance day	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Occupation × Month	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day of month	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day of week	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Holiday	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,096,330	2,709,890	3,323,450	3,630,230	3,067,800	3,834,750	3,630,230	3,834,750
Injuries	2,829	3,647	7,813	14,098	8,420	15,911	10,001	16,081
Clusters	369	477	585	639	540	675	639	675
Eligibility threshold	0.514	0.469	0.308	0.261	0.187	0.134	0.098	0.085
R^2	0.004	0.004	0.010	0.016	0.010	0.019	0.011	0.021

Notes: Coefficients are obtained from estimating equation 1 separately for each gender × age group, relative to days 0–5 of the benefit month, and are linear probability model estimates for any injury, in percentage points. Samples include workers with $\hat{p}$ at or above the worker-weighted median of their gender × age group (eligibility threshold row). All regressions include issuance day, year, occupation × month, day of month, day of week, and holiday fixed effects. Standard errors clustered by issuance day × occupation in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

TABLE A.6—HETEROGENEITY BY AGE AND GENDER: CELL-SPECIFIC MEDIANS (PQML)

	Injury count							
	18–24		25–34		35–44		45–54	
	Female (1)	Male (2)	Female (3)	Male (4)	Female (5)	Male (6)	Female (7)	Male (8)
Days 6–10	−0.0217	−0.0234	0.0137	0.0111	−0.0378	0.0124	0.0429	0.0330
	(0.0732)	(0.0678)	(0.0451)	(0.0323)	(0.0433)	(0.0312)	(0.0426)	(0.0300)
Days 11–15	−0.0893	0.0216	−0.0228	−0.0026	−0.0007	−0.0054	0.0372	0.0207
	(0.0878)	(0.0754)	(0.0502)	(0.0437)	(0.0515)	(0.0373)	(0.0490)	(0.0377)
Days 16–20	−0.2270**	0.2019***	−0.0383	−0.0077	−0.0433	0.0074	0.0612	0.0662*
	(0.0887)	(0.0752)	(0.0559)	(0.0409)	(0.0593)	(0.0375)	(0.0501)	(0.0363)
Days 21–25	−0.2502***	0.1287*	0.0061	0.0109	0.0086	0.0111	0.0559	0.0041
	(0.0919)	(0.0742)	(0.0556)	(0.0388)	(0.0534)	(0.0372)	(0.0488)	(0.0349)
Days 26–30	−0.1715**	0.1288**	0.0369	0.0478	−0.0717	−0.0371	0.0305	−0.0015
	(0.0704)	(0.0635)	(0.0509)	(0.0370)	(0.0451)	(0.0329)	(0.0375)	(0.0302)
Fixed effects								
Issuance day	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Occupation × Month	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day of month	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day of week	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Holiday	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,401,570	1,704,480	2,626,130	3,064,180	2,607,380	3,357,580	3,072,310	3,396,280
Injuries	2,829	3,647	7,813	14,098	8,420	15,911	10,001	16,081
Clusters	369	477	585	639	540	675	639	675
Eligibility threshold	0.514	0.469	0.308	0.261	0.187	0.134	0.098	0.085
Pseudo R^2	0.082	0.086	0.119	0.137	0.122	0.162	0.131	0.165

Notes: Coefficients are obtained from estimating equation 1 separately for each gender × age group, relative to days 0–5 of the benefit month, and are PQML estimates on injury counts with log cell employment as an offset, interpretable as log changes in injury rates. Samples include workers with $\hat{p}$ at or above the worker-weighted median of their gender × age group (eligibility threshold row). All regressions include issuance day, year, occupation × month, day of month, day of week, and holiday fixed effects. Standard errors clustered by issuance day × occupation in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

TABLE A.7—HETEROGENEITY BY SEVERITY

	Injury count							
	18–24		25–34		35–44		45–54	
	Female (1)	Male (2)	Female (3)	Male (4)	Female (5)	Male (6)	Female (7)	Male (8)
Day of receipt × Non-severe	0.0240	0.1230	−0.1333	0.0557	0.0087	0.0036	0.0107	−0.1611
	(0.0929)	(0.0854)	(0.0937)	(0.0691)	(0.0917)	(0.1060)	(0.1323)	(0.1245)
Day of receipt × Severe	−0.0936	0.2176	−0.0461	−0.1455	−0.2337	−0.0056	0.0520	0.3854***
	(0.1787)	(0.1392)	(0.1679)	(0.1449)	(0.2168)	(0.1396)	(0.1979)	(0.1232)
Days 16–30 × Non-severe	−0.0897**	0.1621***	0.0353	−0.0088	−0.0325	−0.0606	0.0865	−0.0273
	(0.0440)	(0.0438)	(0.0430)	(0.0399)	(0.0524)	(0.0487)	(0.0575)	(0.0530)
Days 16–30 × Severe	−0.1187*	0.2260***	−0.0816	−0.0591	−0.1238*	−0.1218**	0.0705	−0.0486
	(0.0648)	(0.0626)	(0.0668)	(0.0605)	(0.0668)	(0.0589)	(0.1018)	(0.0701)
Fixed effects								
Issuance day	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Occupation × Month	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Occupation × Injury type	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day of month	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day of week	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Holiday	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4,105,450	3,817,920	3,487,040	4,218,220	2,566,500	3,572,140	1,905,140	2,308,950
Severe claims	929	1,486	1,074	2,026	1,018	1,560	869	1,206
Non-severe claims	4,147	4,489	4,765	5,539	3,804	3,475	2,482	2,235
Clusters	513	504	477	423	297	351	198	252
Pseudo R^2	0.135	0.140	0.145	0.121	0.116	0.112	0.140	0.095

Notes: Coefficients are PQML estimates of equation 2, estimated separately for each gender × age group, and are interpretable as log changes in injury rates relative to days 1–15 of the benefit month. The sample includes traumatic injuries for workers with $\hat{p} \geq 0.44$. All regressions include log cell employment as an offset. Standard errors clustered by issuance day × occupation in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

the top bin. Third, the score is linear in the distribution, so the score of a mix of injury types equals the case-weighted average of their scores. Aggregation to broader categories and claim-weighted summaries are therefore internally consistent. Fourth, natures, body parts, and nature × body-part combinations are all scored against the same reference distribution, so their scores can be compared directly.

	Days away from work (fitted share)							
	1	2	3–5	6–10	11–20	21–30	31+	Index
Sprains × knee	.084	.076	.138	.117	.131	.080	.373	0.577
Sprains, all parts	.106	.098	.179	.128	.116	.066	.306	0.525
Sprains × finger	.168	.130	.217	.141	.123	.056	.165	0.431
Fractures × leg (non-knee)	.043	.037	.077	.082	.094	.081	.585	0.689
Fractures, all parts	.054	.048	.095	.091	.110	.089	.513	0.655
Fractures × finger	.077	.063	.124	.113	.128	.091	.404	0.598
All injuries (raw)	.140	.109	.174	.118	.110	.064	.285	0.500
All injuries (cum.)	.140	.249	.423	.541	.651	.715	1	–
$\bar{p}_b$	.07	.195	.336	.482	.596	.683	.857	=

Table B.1: Subset of fitted DAFW distribution, pooled 2011-2018

Notes: Distributions are estimated using an iterative proportional fitting (IPF) algorithm to generate observed counts of nature × body part, nature × DAFW, and body part × DAFW from BLS Tables R19, R67, and R68 for 2011 through 2018. The severity index for nature j and part p is calculated as $Index_{jp} = \sum_b w_{jpb}\bar{p}_b$, where w_{jpb} is the fitted share of jp type injuries in DAFW bin b and $\bar{p}_b$ is the midpoint of the *cumulative* national share. It therefore has a direct probability interpretation. $S_j = P(X_j > R) + \frac{1}{2}P(X_j = R)$, where X_j is the duration of a random injury of type j and R that of a random national DAFW injury: the probability that an injury of type j keeps a worker away longer than a typical lost-time injury, with ties split evenly. A type whose durations match the nation's scores exactly 0.5. Scores above 0.5 mean durations longer than the national norm.

	Nature only	Eye	Finger	Hand (non-finger)	Wrist	Shoulder	Lumbar back	Knee	Ankle
Fractures	0.655	(0.371)	0.598	0.631	0.673	0.750	0.669	0.725	0.661
Sprains, strains	0.525	(0.262)	0.431	0.461	0.506	0.609	0.503	0.577	0.489
Soreness, pain	0.497	0.242	0.410	0.442	0.488	0.597	0.487	0.563	0.471
Bruises	0.413	0.223	0.353	0.379	0.416	0.513	0.417	0.483	0.401
Cuts, lacerations	0.373	0.218	0.360	0.388	0.427	(0.529)	(0.429)	0.498	0.412
Part only		0.258	0.428	0.431	0.562	0.611	0.501	0.571	0.516

Table B.2: Combined nature × part severity indices

Notes: This table shows estimated severity indices for a subset of nature × body part combinations based on pooled national data from 2011 to 2018 (BLS, 2018a,b,c). Values shown in parentheses are based on fewer than 1,000 injuries (e.g., 340 cases of sprain × eye). The fact that body parts scale each nature index by the same factor is an algorithmic assumption and should not be taken as evidence that this is in fact true.

C Sizing attenuation bias

My estimates of the SNAP benefit-cycle effect on occupational injuries are subject to two primary sources of measurement error due to the nature of my data. First, I do not observe SNAP participation. The estimation sample includes all workers whose observable characteristics place them above a threshold of predicted SNAP eligibility, so it necessarily pools participants with non-participants, whose injury risk should not respond to the SNAP benefit cycle. I will refer to this as *dilution*. Even in the case of zero dilution, because households in Oregon are assigned issuance days based on the last digit of the primary applicant's SSN, a second source of measurement error stems from not observing the household in my claims data. For each claim, I assign the issuance day implied by the worker's own SSN, which is correct when the worker is the primary applicant and is an independent draw from the issuance-day distribution when the worker is not. I will refer to this as *misallocation*.

To fix ideas, abstract from covariates, weights, and the full set of fixed effects in the main specification and consider the simplified linear model for a particular date t:

$$y_c \;=\; \gamma_{t(c)} \;+\; \sum_{k=1}^{K} \beta_k \, \mathbf{1}\{b_c = k\} \;+\; \varepsilon_c \tag{3}$$

where $c \equiv sdo$ indexes the terminal SSN digit-gender-age-occupation cell, $\gamma_{t(c)}$ is a date effect standing in for the fixed effects in the main specification, and $b_c \in \{0, 1, \ldots, K\}$ is the days-since-issuance bin, with $b = 0$ the omitted reference bin ($\beta_0 = 0$).

Throughout, β_k denotes the causal effect of being k bins from issuance *for a SNAP participant whose issuance day is correctly measured*. All derivations are statements about conditional means and therefore carry over unchanged to the weighted least squares estimates reported in the main text.

Two facts about Oregon's assignment mechanism do most of the work below. SSN last digits are as good as randomly assigned and are independent across individuals, including members of the same household. This assignment mechanism motivates: (1) true issuance days are independent of worker characteristics and of the calendar date of injury, and (2) when a worker is not the primary applicant, the worker's own SSN digit, and thus the implied issuance day, is an independent draw from the schedule, unrelated to the household's true day.

Dilution from unobserved SNAP participation

Bias

Let $S_c \in \{0, 1\}$ indicate the SNAP participation rate for cell c, which is not observed. Since only participants are exposed to the benefit cycle, the outcome equation is then

$$y_c \;=\; \boldsymbol{\gamma}_t \;+\; S_c \sum_k \beta_k \, \mathbf{1}\{b_c = k\} \;+\; \varepsilon_c. \tag{4}$$

For non-participants, the assigned bin is a placebo, as it is constructed from the worker's own SSN digit, but no benefits arrive on any day so it carries no effect.

Define $\pi \equiv \Pr(S_c = 1 \mid c \in \text{sample})$ as the participation share among cells in the analysis sample.

Assumption A1 (Participation is orthogonal to the assigned bin)**.** $\Pr(S_c = 1 \mid b_c = k,\, t) = \pi$ *for all* k *and* t*.*

Assumption A1 is substantiated by random terminal digit assignment. SNAP participation is determined by income, household structure, and take-up, none of which depend on the last digit of an SSN. Though the sample was selected based on *predicted eligibility*, this does not threaten assumption A1 as the variables used to estimate eligibility are independent of the assigned bin. Rather, it only determines the level of π.

Taking expectations of (4) conditional on the assigned bin and date, and applying Assumption A1,

$$\begin{aligned} E[y_c \mid b_c = k,\, t] &= \boldsymbol{\gamma}_t + \Pr(S_c = 1 \mid b_c = k,\, t)\, \beta_k \\ &= \boldsymbol{\gamma}_t + \pi\, \beta_k. \end{aligned} \tag{5}$$

A saturated regression of y on bin indicators and date effects recovers differences in these conditional means, so the probability limit of the estimated coefficient on bin k is

$$\hat{\beta}_k \;\xrightarrow{p}\; \big(\boldsymbol{\gamma}_t + \pi\beta_k\big) - \big(\boldsymbol{\gamma}_t + \pi\beta_0\big) \;=\; \pi\, \beta_k. \tag{6}$$

Dilution therefore attenuates every bin coefficient by the same proportional factor π. Because the factor is common across bins, the *shape* of the estimated profile is unbiased; only its scale is compressed. Equation (6) is the standard intention-to-treat result - my estimates are reduced-form effects of *predicted-eligible*

status on injury outcomes, and $\beta_k = \hat{\beta}_k/\pi$ recovers the effect for participants.

Implied magnitudes and implications for inference

The participation share decomposes as $\pi = \tau\bar{p}$, where $\bar{p}$ is the mean probability of SNAP eligibility among retained cells and τ is take-up conditional on eligibility. At the median threshold, each worker in cell c satisfies $p_i \geq 0.22$, so $\bar{p}$ is bounded-below at 0.22 and the worker-weighted average across above-median cells in my sample is roughly 0.45. For take-up, FNA estimates place Oregon's SNAP participation rate among the eligible working-poor population over the study period near 90 percent, so $\tau \approx 0.9$. These values imply that $\pi \approx 0.40$, so the true effect for participants in bin k is expected to be $1/0.40 = 2.5$ times the estimated coefficient.

Dilution leaves the variance of the estimator essentially unchanged while shrinking its expectation. To see this, decompose the conditional variance of the outcome under (4):

$$\mathrm{Var}(y_c \mid b_c = k,\, t) \;=\; \pi(1-\pi)\,\beta_k^2 \;+\; \sigma_\varepsilon^2, \tag{7}$$

where the first term reflects heterogeneity between participants and non-participants within a bin. In the binary outcome case, β_k is on the order of 10^{-4} while σ_ε^2 is on the order of the outcome variance itself, so $\pi(1-\pi)\beta_k^2$ is smaller than σ_ε^2 by several orders of magnitude and the residual variance is unaffected to first order. The regressors are similarly unchanged, because assigned bins are constructed identically for participants and non-participants. It follows that

$$\mathrm{se}\left(\hat{\beta}_k\right) \;\approx\; \text{unchanged}, \qquad \frac{\hat{\beta}_k}{\mathrm{se}(\hat{\beta}_k)} \;\approx\; \pi \times \frac{\beta_k}{\mathrm{se}(\hat{\beta}_k)}. \tag{8}$$

Under the null $\beta_k = 0$, the diluted effect $\pi\beta_k$ is also zero, so conventional tests will be correctly sized, but the t-statistic against any fixed alternative is deflated by π. Thus the minimum detectable effect for *true participants* is higher by a factor of $1/\pi$. This in part explains the width of my confidence intervals relative to studies with administrative participation data. Finally, note that rescaling the point estimate by $1/\pi$ rescales its confidence interval by the same factor, so while the correction changes the economic magnitude implied by the point estimate, it does not artificially manufacture statistical precision.

Misallocation of issuance days

Bias

Consider now the subpopulation of participants, and suppose worker i in cell c truly receives benefits on day d_i^*, which is unobserved and dependent on the terminal digit of their SNAP household head's SSN. Let $m_c \equiv \Pr(\text{worker } i \text{ is not the SNAP household head} \mid c(i) = c)$. By independence of SSN digits across household members,

$$d_i^* = \begin{cases} d_i & \text{with probability } 1 - m_c, \\ \tilde{d} \sim p(\cdot), & \text{with probability } m_c, \end{cases} \tag{9}$$

where d_i is the issuance day implied by i's terminal SSN digit, and is $p(\cdot)$ is the population issuance-day distribution. Under Oregon's schedule, $p_1 = 0.2$ for the first of the month, which pools SSN digits 0 and 1, and $p_d = 0.1$ for $d = 2...9$.

Bins are a deterministic function of the calendar date and the issuance day, $b = b(t, d)$, where the mapping varies with t. Define the date-specific bin distribution induced by the SNAP schedule,

$$s_j(t) \equiv \sum_d p(d)\, \mathbf{1}\{b(t,d) = j\}, \tag{10}$$

which is the probability that an independent draw from $p(d)$ lands in bin j for a worker injured on date t.

The object in (10) is common to all workers injured on date t, regardless of issuance day, and this property is the foundation for the derivations that follow. From (9), the distribution of the true bin conditional on the observed bin and the date is

$$\Pr\left(b_i^* = j \mid b_i = k,\, t\right) = (1 - m_c)\, \mathbf{1}\{j = k\} + m_c\, s_j(t). \tag{11}$$

Since outcomes depend on the *true* bin, taking expectations of the participant analogue of (3) conditional on

the observed bin yields:

$$
\begin{aligned}
E[y_i \mid b_i = k,\, t,\, c] &= \boldsymbol{\gamma}_t + \sum_j \Pr\left(b_i^* = j \mid b_i = k,\, t\right) \beta_j \\
&= \boldsymbol{\gamma}_t + (1 - m_c)\, \beta_k + m_c \sum_j s_j(t)\, \beta_j \\
&= \boldsymbol{\gamma}_t + (1 - m_c)\, \beta_k + m_c\, \bar{\beta}(t), \qquad \bar{\beta}(t) \equiv \sum_j s_j(t)\, \beta_j,
\end{aligned} \tag{12}
$$

where $\bar{\beta(t)}$ is the expected bin effect for participating workers injured on date t if their true issuance days were randomly drawn from the population distribution. Note that the contamination term $m_c\, \bar{\beta}(t)$ does not depend on k. Because the redraw in (9) is independent of d_i, each bin is identically contaminated conditional on date t. Differencing (12) against the reference bin *within a date*, the contamination term cancels:

$$
E[y_i \mid b_i = k, t, c] - E[y_i \mid b_i = 0, t, c] \;=\; (1 - m_c)\, \beta_k. \tag{13}
$$

Thus, as (13) holds on every date, any weighted average across dates equals $(1 - m_c)\beta_k$ as well, and a specification with full date effects would yield estimates proportionately attenuated by a factor of $1 - m_c$. Since my specification only includes year, month, day-of-month, day-of-week, and holiday fixed effects rather than full date effects, the linear combination of these dummy variables spans $\bar{\beta}(t)$ closely but not perfectly. Therefore, $1 - m_c$ should be viewed as a close approximation for the true misallocation bias factor.

Implied magnitudes and implications for inference

Mirroring the main text, let d index the unique combinations of gender and age bin in my sample. I estimate m_d using the sample of working SNAP participants in Oregon observed in the SNAP Quality Control database over my study period to proxy for the expected misallocation rate for gender-age-*occupation* cells (m_c) in my analysis. The QC data include the gender, age, and employment status for all members of the SNAP household, but not workers' occupations, so if head-of-household shares among d workers differed systematically across occupations, the m_d estimate will be inflated or deflated relative to the relevant m_c for cells in my sample. Samples in the QC data are constructed to be representative of SNAP participants at the state-level so by extension, the sampled *workers* should be representative of working SNAP participants in

Oregon. Thus, m_d can be interpreted as an estimate of the average misallocation rate for d workers across all occupations.

Table C.1: Misallocation rates and implied corrections by gender and age

d	Headship rate (QC data)	Misallocation m_d	Attenuation $\lambda_d = 1 - m_d$	Correction $1/\lambda_d$
Women, 18–24	0.85	0.15	0.85	1.18
Women, 25–34	0.94	0.06	0.94	1.06
Women, 35–44	0.92	0.08	0.92	1.09
Women, 45–54	0.92	0.08	0.92	1.09
Men, 18–24	0.63	0.37	0.63	1.59
Men, 25–34	0.64	0.36	0.64	1.56
Men, 35–44	0.62	0.38	0.62	1.61
Men, 45–54	0.77	0.23	0.77	1.30

Note: These estimates are based on the author's own analysis of the SNAP Quality Control database for fiscal years 2005 to 2018. The number of d workers in each household are aggregated using FNA household weights.

Table C.1 reports the derived attenuation factors $\lambda_d = 1 - m_d$ and implied estimate corrections $1/\lambda_d$ for each group d. Two features of table C.1 directly affect the interpretation of the heterogeneity in figure 3 from the main text. First, attenuation differs drastically by gender due to headship rates being higher among female workers across age bins. For example, if true effects were identical for SNAP participants age 35 to 44 regardless of gender, my estimates for men would appear $1 - 0.62/0.92 \approx 33$ percent smaller than the estimates for women in this age group purely due to issuance day misallocation.

Second, attenuation differs across age bins within gender, though less dramatically than the cross-gender comparisons. For men, headship rates are highest among workers age 45 to 54, so in the case of identical true bin effects, the estimates for men age 35 to 44 would appear $1 - 0.62/0.77 \approx 19$ percent smaller than those for men in the oldest group. Conversely, I find that 94 percent of female workers aged 25 to 34 are their SNAP household heads, compared to only 85 percent of workers under age 25, so identical true effects would appear $1 - 0.85/0.94 \approx 10$ percent smaller for the age 18 to 24 workers relative to 25-34 workers.

As with dilution, misallocation will affect the conditional means by amounts of order $\beta \approx 10^{-4}$, which are negligible relative to the residual standard deviation. The standard error of $\hat{\beta}_k$ is therefore largely unaffected by m_c, so the t-statistic scales with the attenuation factor $(1-m_c)$. Measurement error in true issuance days thus costs statistical power proportionally with the attenuation factor and leaves test size unaffected, by the same argument as in the dilution case.

However, unlike in the dilution case, misallocation will contaminate within-cluster correlation in a way that shrinks standard errors clustered at this level. Suppose residual dependence operates at the level of the true issuance day:

$$\mathrm{Cov}(u_i, u_j) = \rho\,\sigma^2\,\mathbf{1}\{d_i^* = d_j^*\}, \qquad i \neq j, \tag{14}$$

while standard errors are clustered on the *assigned* day based on the observed terminal digit. Workers sharing a true issuance day may be misallocated, which would violate the core assumption that residual dependence exists within but not across clusters. Furthermore, within-cluster covariance would be diluted since workers within a cluster share a true day only probabilistically. For two observations assigned to day d, independence of the redraws across workers gives

$$\begin{aligned}\Pr\left(d_i^* = d_j^* \mid d_i = d_j = d\right) &= \left[(1-m) + m\,p_d\right]^2 + \sum_{d' \neq d} \left(m\,p_{d'}\right)^2 \\ &\approx \left(1 - 0.9\,m\right)^2,\end{aligned} \tag{15}$$

where the approximation uses $p_d \approx 0.1$ and drops the second term, which is at most $0.11\,m^2$ (about two percent of the first term at $m = 0.31$). Expression (15) is the share of the true within-cluster covariance that survives into the estimator, which is approximately 0.88 for women assuming a misallocation rate $m = 0.07$ and 0.52 for men assuming a misallocation rate of $m = 0.31$. Misallocation therefore attenuates measured within-cluster dependence, though not completely. If headship rates were below 0.33, so $m > 0.68$, this would eradicate 85 percent of the within-cluster covariance, but tabulations based on the QC data suggest this is not the case for any group of workers in my sample.

Two considerations bound the practical importance of this channel. First, the economic content of dependence at the true issuance-day level is the benefit cycle itself, and the specification models that cycle directly through the bin indicators and the issuance-day fixed effects; the residual can carry only the component of the cycle response that the bin partition fails to capture.[20] Second, dilution compounds the bound: with participation share π, only participant pairs can share a true benefit cycle, scaling any residual dependence by a further factor of order π^2. Consistent with both considerations, comparisons of cluster-robust to heteroskedasticity-robust variance estimates across candidate clustering dimensions (available upon request)

[20] One caveat to this is that since the issuance-day fixed effects are defined on *assigned* days, the true-day common component for misallocated workers is not captured, and this component is m-weighted and second order.

show no detectable within-cluster dependence on any dimension with enough clusters for the comparison to be informative.

Combined correction

The two error processes compound multiplicatively. Misallocation operates only within the participant subpopulation, so combining (5) and (12), the conditional mean in cell c is

$$E[y_i \mid b_i = k,\, t,\, c] = \boldsymbol{\gamma}_t + \pi_c \Big[(1 - m_c)\, \beta_k + m_c\, \bar{\beta}(t) \Big], \tag{16}$$

and the within-date contrast identifies

$$\hat{\beta}_k \xrightarrow{p} \pi_c\, (1 - m_c)\, \beta_k \tag{17}$$

in cell c, with the pooled estimate attenuated by the corresponding weighted average. With $\pi \approx 0.40$ and the estimates of misallocation rates m_d in Table C.1, the combined attenuation factor $\pi\lambda_c$ ranges from $0.40 \times 0.94 \approx 0.38$ (women 25–34) to $0.40 \times 0.62 \approx 0.25$ (men 35–44): the estimated coefficients understate per-participant effects by a factor of roughly 2.7 to 4. Dilution accounts for the majority of the gap, while misallocation affects the variation in magnitudes across demographic cells.

References

ARTEAGA, I., C. HEFLIN, AND L. HODGES (2018): "SNAP benefits and pregnancy-related emergency room visits," *Population Research and Policy Review*, 37, 1031–1052.

BEATTY, T. AND J. WEILL (2024): "Social Security and High-Frequency Labor Supply: Evidence from Uber Drivers," .

BEATTY, T. K., M. P. BITLER, X. H. CHENG, AND C. VAN DER WERF (2019): "SNAP and paycheck cycles," *Southern Economic Journal*, 86, 18–48.

BLS (2018a): "Table R19. Number of Nonfatal Occupational Injuries and Illnesses Involving Days Away from Work by Part of Body and Selected Natures of Injury or Illness, Private Industry," Survey of Occupational Injuries and Illnesses, Case and Demographic Characteristics Resource Tables, corrected estimates for reference year 2011.

——— (2018b): "Table R67. Number of Nonfatal Occupational Injuries and Illnesses Involving Days Away from Work by Nature of Injury or Illness and Number of Days Away from Work, and Median Number of Days Away from Work, Private Industry," Survey of Occupational Injuries and Illnesses, Case and Demographic Characteristics Resource Tables, corrected estimates for reference year 2011.

——— (2018c): "Table R68. Number of Nonfatal Occupational Injuries and Illnesses Involving Days Away from Work by Part of Body and Number of Days Away from Work, and Median Number of Days Away from Work, Private Industry," Survey of Occupational Injuries and Illnesses, Case and Demographic Characteristics Resource Tables.

BOND, T. N., J. B. CARR, A. PACKHAM, AND J. SMITH (2022): "Hungry for success? SNAP timing, high-stakes exam performance, and college attendance," *American Economic Journal: Economic Policy*, 14, 51–79.

BOONE, J. AND J. C. VAN OURS (2006): "Are recessions good for workplace safety?" *Journal of Health economics*, 25, 1069–1093.

BOONE, J., J. C. VAN OURS, J.-P. WUELLRICH, AND J. ZWEIMÜLLER (2011): "Recessions are bad for workplace safety," *Journal of health economics*, 30, 764–773.

BRONCHETTI, E. T. AND M. MCINERNEY (2012): "Revisiting incentive effects in workers' compensation: Do higher benefits really induce more claims?" *ILR Review*, 65, 286–315.

BROSS, I. D. (1958): "How to use ridit analysis," *Biometrics*, 18–38.

CABRAL, M., C. CUI, AND M. DWORSKY (2022): "The demand for insurance and rationale for a mandate: Evidence from workers' compensation insurance," *American Economic Review*, 112, 1621–1668.

CABRAL, M. AND M. DILLENDER (2024): "Air pollution, wildfire smoke, and worker health," Tech. rep., National Bureau of Economic Research.

CARD, D. AND B. P. MCCALL (1996): "Is workers' compensation covering uninsured medical costs? Evidence from the "monday effect"," *ILR Review*, 49, 690–706.

CARR, J. B. AND A. PACKHAM (2019): "SNAP benefits and crime: Evidence from changing disbursement schedules," *Review of Economics and Statistics*, 101, 310–325.

COOK, J. AND C. N. EAST (2026): "The disenrollment and labor market effects of SNAP work requirements on parents," *Journal of Public Economics*, 257, 105631.

COTTI, C., J. GORDANIER, AND O. OZTURK (2018): "When does it count? The timing of food stamp receipt and educational performance," *Economics of Education Review*, 66, 40–50.

COTTI, C. D., J. M. GORDANIER, AND O. D. OZTURK (2020): "Hunger pains? SNAP timing and emergency room visits," *Journal of health economics*, 71, 102313.

CUBBIN, C., F. B. LECLERE, AND G. S. SMITH (2000): "Socioeconomic status and the occurrence of fatal and nonfatal injury in the United States," *American journal of public health*, 90, 70.

CUNNYNGHAM, K. (2021): "Reaching Those in Need: Estimates of State Supplemental Nutrition Assistance Program Participation Rates in 2018," Prepared by Mathematica for the Food and Nutrition Service under Contract No. 12-3198-20-F-0051.

DAVIES, M., R. J. PARK, AND A. STANSBURY (2026): "How Do Minimum Wages Affect Workplace Injuries?" *Employment Research Newsletter*, 33, 1.

DEMING, W. E. AND F. F. STEPHAN (1940): "On a least squares adjustment of a sampled frequency table when the expected marginal totals are known," *The Annals of Mathematical Statistics*, 11, 427–444.

DILLENDER, M. (2015): "The effect of health insurance on workers' compensation filing: Evidence from the affordable care act's age-based threshold for dependent coverage," *Journal of Health Economics*, 43, 204–228.

——— (2021): "Climate change and occupational health: Are there limits to our ability to adapt?" *Journal of Human Resources*, 56, 184–224.

FNA (2026): "Supplemental Nutrition Assistance Program (SNAP) Quality Control Database, Fiscal Years 2005–2018," Data files hosted by Mathematica under contract with USDA FNA, accessed: 2026-10-05.

FNS (2025): "SNAP Program Access Index," Online database, page updated January 24, 2025; accessed via SNAP Quality Control.

GASSMAN-PINES, A. AND L. BELLOWS (2018): "Food instability and academic achievement: A quasi-experiment using SNAP benefit timing," *American Educational Research Journal*, 55, 897–927.

GENNETIAN, L. A., R. SESHADRI, N. D. HESS, A. N. WINN, AND R. M. GOERGE (2016): "Supplemental Nutrition Assistance Program (SNAP) benefit cycles and student disciplinary infractions," *Social Service Review*, 90, 403–433.

HAMRICK, K. S. AND M. ANDREWS (2016): "SNAP participants' eating patterns over the benefit month: a time use perspective," *PloS one*, 11, e0158422.

HRADIL, V. (2018): "Does minimum wage affect workplace safety?" *CERGE-EI Working Paper Series*.

HU, Z. AND Z. WANG (2019): "Nutrition, labor supply, and productivity: Evidence from ramadan in indonesia," *Available at SSRN 3450834*.

JONES, J. W. (2025): "Supplemental nutrition assistance program (SNAP)–key statistics and research," *USDA ERS*.

KANTOR, S., W. WINKELSTEIN JR, AND M. A. IBRAHIM (1968): "A note of the interpretation of the ridit as a quantile rank," *American Journal of Epidemiology*, 87, 609–615.

KRUEGER, A. B. (1990a): "Incentive effects of workers' compensation insurance," *Journal of Public Economics*, 41, 73–99.

——— (1990b): "Workers' compensation insurance and the duration of workplace injuries," .

KUHN, M. A. (2018): "Who feels the calorie crunch and when? The impact of school meals on cyclical food insecurity," *Journal of Public Economics*, 166, 27–38.

LAFIANDRA, W. AND D. SCHWAB (2023): "A higher minimum wage promotes workplace safety: Evidence from the restaurant industry," in *Proceedings of The 7th International Conference on Applied Research in Management, Economics and Accounting (94-114). https://www. doi. org/10.33422/7th. iarmea*, vol. 125.

LEE, G. AND T. BEATTY (2025): "Wildfires and agricultural worker injury," *American Journal of Agricultural Economics*.

LEE, J. Y., R. M. NAYGA JR, Y. JO, AND B. J. RESTREPO (2022): "Time use and eating patterns of SNAP participants over the benefit month," *Food Policy*, 106, 102186.

MARKS, M., S. PRINA, AND R. TAHAJ (2024): "Short-term labor supply response to the timing of transfer payments: Evidence from the SNAP program," *Labour Economics*, 91, 102636.

MEN, F., M. L. URQUIA, AND V. TARASUK (2021): "Examining the relationship between food insecurity and causes of injury in Canadian adults and adolescents," *BMC public health*, 21, 1557.

MORENO, F. J., B. BIGELOW, AND M. A. JOHNSON (2025): "Working on an empty stomach: How food insecurity impacts job performance through rumination," *Personnel Psychology*, 78, 794–813.

MOY, J. H., U. A. KHAN, W. J. ONG, AND C. M. BARNES (2026): "The effects of food insecurity on work outcomes." *Journal of Applied Psychology*.

NASI (2024): "Workers' Compensation: Benefits, Costs, and Coverage, 2022 Data," Tech. rep., National Academy of Social Insurance, 27th annual report.

NSC (2026): "Work Injury Costs," https://injuryfacts.nsc.org/work/costs/work-injury-costs/, accessed on February 20, 2026; estimates of the total cost of work injuries in the United States via the Injury Facts database.

ROSENBAUM, D., J. LLOBRERA, C. NCHAKO, AND L. NUÑEZ (2026): "SNAP Tracker: People Are Losing Food Assistance as the Harmful 2025 Republican Reconciliation Law Is Implemented," Center on Budget and Policy Priorities, text last updated July 21, 2026.

RUGGLES, S., S. FLOOD, M. SOBEK, D. BACKMAN, G. COOPER, J. A. RIVERA DREW, S. RICHARDS, R. RODGERS, J. SCHROEDER, AND K. C. W. WILLIAMS (2025): "IPUMS USA: Version 16.0 [dataset]," .

SELIGMAN, H. K., A. F. BOLGER, D. GUZMAN, A. LÓPEZ, AND K. BIBBINS-DOMINGO (2014): "Exhaustion of food budgets at month's end and hospital admissions for hypoglycemia," *Health affairs*, 33, 116–123.

SHAPIRO, J. M. (2005): "Is there a daily discount rate? Evidence from the food stamp nutrition cycle," *Journal of public Economics*, 89, 303–325.

SMITH, G., G. S. SOROCK, H. WELLMAN, T. K. COURTNEY, AND G. S. PRANSKY (2006): "Blurring the distinctions between on and off the job injuries: similarities and differences in circumstances," *Injury Prevention*, 12, 236–241.

SMITH, R. S. (1990): "Mostly on Monday: is workers' compensation covering off-the-job injuries?" in *Benefits, costs, and cycles in workers' compensation*, Springer, 115–127.

STEEGE, A. L., S. L. BARON, S. M. MARSH, C. C. MENÉNDEZ, AND J. R. MYERS (2014): "Examining occupational health and safety disparities using national data: A cause for continuing concern," *American journal of industrial medicine*, 57, 527–538.

STEENLAND, K., W. HALPERIN, S. HU, AND J. T. WALKER (2003): "Deaths due to injuries among employed adults: the effects of socioeconomic class," *Epidemiology*, 14, 74–79.

THOMAS, D., E. FRANKENBERG, J. FRIEDMAN, J.-P. HABICHT, M. HAKIMI, N. INGWERSEN, N. JONES, C. MCKELVEY, G. PELTO, B. SIKOKI, ET AL. (2006): "Causal effect of health on labor market outcomes: Experimental evidence," .

WILDE, P. E. AND C. K. RANNEY (2000): "The monthly food stamp cycle: shopping frequency and food intake decisions in an endogenous switching regression framework," *American journal of agricultural economics*, 82, 200–213.